

Stochastic Processes

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is one fixed probability space. One picture carries the lecture: a **table of sample paths**, rows the outcomes ω , columns the times t , the entry $X_t(\omega)$.

- | | |
|------------------------------|---|
| 1. Stochastic process | A family of random variables indexed by time. Rows are paths, columns are random variables. |
| 2. Random walk | Sum of independent ± 1 steps. Reflection, and first-step analysis as the recurring tool. |
| 3. Brownian motion | The random walk seen from far away; $\sqrt{n}$ scaling; independent Gaussian increments. |
| 4. Risk of ruin | A gambler against a casino, solved completely by first-step analysis. |
| 5. Markov chains | The transition matrix is a table whose rows are conditional distributions. Powers, stationary distributions, classes, hitting times, convergence. |

Stochastic Process: Definition

Definition

Let T be a set of **times**, either $T = \{0, 1, 2, \dots\}$ (discrete time) or $T = [0, \infty)$ (continuous time). A **stochastic process** is a family

$$(X_t)_{t \in T}, \quad X_t: \Omega \rightarrow \mathbb{R} \text{ a random variable for each } t \in T,$$

all defined on the same $(\Omega, \mathcal{F}, \mathbb{P})$. Equivalently it is one function of two arguments, $X: T \times \Omega \rightarrow \mathbb{R}$, $(t, \omega) \mapsto X_t(\omega)$.

One random variable per instant of time, all living on the same outcome ω .

- **For fixed** ω , the function $T \rightarrow \mathbb{R}$, $t \mapsto X_t(\omega)$, is the **sample path** (or trajectory) of the outcome ω : an ordinary sequence or an ordinary function of time.
- **For fixed** t , the function $\Omega \rightarrow \mathbb{R}$, $\omega \mapsto X_t(\omega)$, is the random variable X_t : what the process shows at time t .
- The **state space** S is the set of values the X_t take, so $X_t: \Omega \rightarrow S$: usually $S \subseteq \mathbb{R}$ ($S = \{-1, +1\}, \mathbb{Z}, \mathbb{R}$); for Markov chains a finite or countable set of named states.
- Shorthand, used throughout: for $A \subseteq S$, $\{X_t \in A\}$ is the event $\{\omega \in \Omega : X_t(\omega) \in A\}$ and $\mathbb{P}(X_t \in A) = \mathbb{P}(\{\omega \in \Omega : X_t(\omega) \in A\})$; $\mathbb{P}(X_s = x, X_t = y)$ is the probability of the intersection of the two such events.

The Central Picture: A Table of Sample Paths

Three fair coin tosses. $\Omega = \{H, T\}^3$, each ω having probability $\frac{1}{8}$. Put $\xi_t: \Omega \rightarrow \{-1, +1\}$, $\xi_t(\omega) = +1$ if toss t of ω is H and -1 if T, and

$$S_t = \xi_1 + \cdots + \xi_t, \quad S_0 = 0, \quad t \in \{0, 1, 2, 3\}.$$

- A **row** is one outcome ω and shows its whole path $t \mapsto S_t(\omega)$.
- A **column** is one time t and shows the random variable S_t : its entries, with the row probabilities, are the distribution of S_t .
- An event is a set of rows: $\{S_3 = 1\}$ is rows 2, 3, 5, so $\mathbb{P}(S_3 = 1) = \frac{3}{8}$.

Every statement about a process is a statement about this table: a path property reads along a row, a distributional property reads down a column.

ω	$\mathbb{P}(\{\omega\})$	S_0	S_1	S_2	S_3
HHH	$\frac{1}{8}$	0	1	2	3
HHT	$\frac{1}{8}$	0	1	2	1
HTH	$\frac{1}{8}$	0	1	0	1
HTT	$\frac{1}{8}$	0	1	0	-1
THH	$\frac{1}{8}$	0	-1	0	1
THT	$\frac{1}{8}$	0	-1	0	-1
TTH	$\frac{1}{8}$	0	-1	-2	-1
TTT	$\frac{1}{8}$	0	-1	-2	-3

rows: paths $t \mapsto S_t(\omega)$; columns: random variables S_t

Reading the Table Two Ways, and What a Process Is Not

Row questions (about paths)

Does the path ever reach 2? Does it return to 0? What is its maximum? Each is a set of rows, hence an event: $\{\max_{t \leq 3} S_t \geq 2\}$ is rows 1, 2, probability $\frac{2}{8}$.

Column questions (about one time)

What is $\mathbb{E}[S_2]$? $\mathbb{P}(S_3 > 0)$? Read column 2 or 3 against the row probabilities:

$$\mathbb{E}[S_2] = \frac{1}{8}(2 + 2 + 0 + 0 + 0 + 0 - 2 - 2) = 0,$$
$$\mathbb{P}(S_3 > 0) = \frac{4}{8}.$$

Mixed questions (about several columns)

$\mathbb{P}(S_1 = 1, S_3 = 1)$ is rows 2, 3: $\frac{2}{8}$. Different columns are *not* independent in general:

$$\mathbb{P}(S_1 = 1)\mathbb{P}(S_3 = 1) = \frac{1}{2} \cdot \frac{3}{8} \neq \frac{2}{8}.$$

A stochastic process is **not**:

- a single random variable: it is one per time;
- a list of distributions, one per column: the distributions of the S_t separately do not determine how columns move together, i.e. the joint distribution of the row;
- a fixed function of time: the path depends on ω , and different rows are different functions.

The object is the whole table: its columns are random variables and its rows are the possible histories.

The Distribution of a Process

Definition

The **finite-dimensional distributions** of $(X_t)_{t \in T}$ are the joint distributions of $(X_{t_1}, \dots, X_{t_n})$ for all n and all times $t_1 < \dots < t_n$ in T : in the discrete case the joint pmfs

$$p_{X_{t_1}, \dots, X_{t_n}} : \mathbb{R}^n \rightarrow [0, 1], \quad (x_1, \dots, x_n) \mapsto \mathbb{P}(X_{t_1} = x_1, \dots, X_{t_n} = x_n).$$

Pick any finite set of columns; the joint distribution of those columns is what the model must specify, and these joint distributions together determine every probability about the process.

Definition (independence)

Random variables $X_1, \dots, X_n$ are **independent**, written $X_1 \perp\!\!\!\perp X_2 \perp\!\!\!\perp \dots \perp\!\!\!\perp X_n$, if for all sets $A_1, \dots, A_n \subseteq \mathbb{R}$

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(X_1 \in A_1) \cdots \mathbb{P}(X_n \in A_n).$$

A sequence $(X_n)_{n \geq 1}$ is independent if every finite sub-family is.

- In the table, the coin tosses ξ_1, ξ_2, ξ_3 are independent, but the partial sums S_1, S_2, S_3 are not (previous slide). A model is usually specified through independent building blocks and a rule, not by writing down all joint pmfs.

The Bernoulli Process and the Simple Random Walk

Definition

Fix $p \in (0, 1)$ and put $q = 1 - p$. A **Bernoulli process** is an independent sequence $\xi_1, \xi_2, \dots$ of random variables $\xi_n: \Omega \rightarrow \{-1, +1\}$ with $\mathbb{P}(\xi_n = +1) = p$, $\mathbb{P}(\xi_n = -1) = q$ for every n . The **simple random walk** is the process $(S_n)_{n \geq 0}$,

$$S_n: \Omega \rightarrow \mathbb{Z}, \quad S_0 = 0, \quad S_n = \xi_1 + \xi_2 + \dots + \xi_n.$$

It is **symmetric** (or fair) if $p = \frac{1}{2}$.

Toss a coin forever and keep a running score, +1 per head and -1 per tail; S_n is the score after n tosses.

- The table on slide 3 is this process up to $n = 3$. Each row is a path $n \mapsto S_n(\omega)$ that moves by exactly ± 1 each step.
- **Independent increments:** for $m < n$, $S_n - S_m = \xi_{m+1} + \dots + \xi_n$ depends only on tosses after m , so it is independent of $(S_0, \dots, S_m)$ and has the same distribution as S_{n-m} . The walk restarts from wherever it is.

Distribution of S_n

Theorem

Let $H_n: \Omega \rightarrow \{0, \dots, n\}$, $H_n = \sum_{k=1}^n \mathbf{1}\{\xi_k = +1\}$, be the number of $+1$ steps among the first n : a sum of n independent Bernoulli(p) indicators, hence binomial(n, p), and $S_n = H_n - (n - H_n) = 2H_n - n$. Hence, for $k \in \{-n, -n+2, \dots, n\}$,

$$\mathbb{P}(S_n = k) = \binom{n}{\frac{n+k}{2}} p^{(n+k)/2} q^{(n-k)/2}, \quad \mathbb{E}[S_n] = n(p - q), \quad \text{Var}(S_n) = 4npq.$$

S_n is a shifted and stretched binomial, so its mean grows like n and its spread like $\sqrt{n}$.

Proof

$S_n = k$ iff $H_n = \frac{n+k}{2}$, which requires $n+k$ even. $\mathbb{E}[S_n] = 2\mathbb{E}[H_n] - n = 2np - n = n(p - q)$ and $\text{Var}(S_n) = 4 \text{Var}(H_n) = 4npq$. $\square$

Fair walk, $n = 4$

k	-4	-2	0	2	4
$\mathbb{P}(S_4 = k)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$\mathbb{E}[S_4] = 0$, $\text{Var}(S_4) = 4$, standard deviation $2 = \sqrt{4}$.

A Path Question: Does the Walk Reach 3 Within 10 Steps?

Fair walk. What is $\mathbb{P}(\max_{k \leq 10} S_k \geq 3)$? This is a row question: it depends on the whole path, not on one column.

Head-on

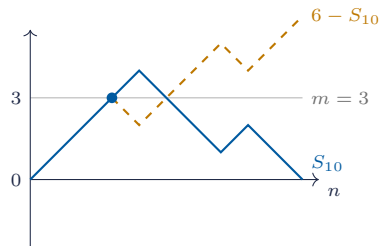
Count the $2^{10} = 1024$ paths that touch level 3 at some time. There is no column to read; one has to track the running maximum along each row.

Theorem (reflection principle, fair walk)

For integers $m \geq 1$ and $n \geq 1$,

$$\mathbb{P}\left(\max_{k \leq n} S_k \geq m\right) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m).$$

A path question about the maximum is turned into a column question about S_n alone.



After the first visit to m , reflect the rest of the path in the line m . This pairs each path that reaches m and ends below m with one ending above m , and the two have equal probability.

Reflection Principle: Proof

Proof

Let $A = \{\max_{k \leq n} S_k \geq m\}$. Split A by where the path ends:

$$\mathbb{P}(A) = \mathbb{P}(A, S_n > m) + \mathbb{P}(A, S_n = m) + \mathbb{P}(A, S_n < m).$$

The first two events do not need A : a path ending at or above m has certainly reached m , so they equal $\mathbb{P}(S_n > m)$ and $\mathbb{P}(S_n = m)$.

For the third, let τ be the first time the path hits m . Map each path in $\{A, S_n < m\}$ to the path that agrees with it up to time τ and thereafter takes the opposite steps. The new path ends at $m + (m - S_n) > m$, and it has the same number of steps, each of probability $\frac{1}{2}$, so the same probability 2^{-n} . The map is a bijection onto $\{S_n > m\}$ (reverse it by reflecting after the first visit to m). Hence $\mathbb{P}(A, S_n < m) = \mathbb{P}(S_n > m)$. $\square$

The fair walk is a uniform distribution on 2^n rows, so probabilities are counts, and reflection is a counting bijection between rows.

- For $p \neq \frac{1}{2}$ the reflected path has a different probability, and the identity fails; the ideas survive as inequalities.
- Brownian motion inherits the statement in the form $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$ (slide 23).

Return to the Problem: Reaching 3 Within 10 Steps

The answer

S_{10} is even, so $\mathbb{P}(S_{10} = 3) = 0$, and

$$\mathbb{P}(S_{10} > 3) = \mathbb{P}(S_{10} \geq 4) = \mathbb{P}(H_{10} \geq 7) = \frac{120 + 45 + 10 + 1}{1024} = \frac{11}{64}. \text{ Hence}$$

$$\mathbb{P}\left(\max_{k \leq 10} S_k \geq 3\right) = 2 \cdot \frac{11}{64} + 0 = \frac{11}{32} \approx 0.34.$$

A brute-force count of the 1024 paths gives 352, agreeing.

Corollary: the distribution of the maximum

With $M_n = \max_{k \leq n} S_k$ and the theorem applied at m and $m + 1$,

$$\mathbb{P}(M_n = m) = \mathbb{P}(M_n \geq m) - \mathbb{P}(M_n \geq m + 1) = \mathbb{P}(S_n = m) + \mathbb{P}(S_n = m + 1) \quad (m \geq 0).$$

The whole distribution of a path quantity has been read off one column of the table.

- Fair walk, $n = 10$: $\mathbb{P}(M_{10} = 0) = \mathbb{P}(S_{10} = 0) + \mathbb{P}(S_{10} = 1) = \frac{252}{1024}$, so about a quarter of all 10-step paths never rise above their start.

First-Step Analysis: Expected Time to Leave $(-a, a)$

Fair walk started at $i \in \{-a+1, \dots, a-1\}$, position $X_n = i + S_n$. Write $\mathbb{P}_i(\cdot)$, $\mathbb{E}_i[\cdot]$ for probabilities and expectations when the start is i (the notation serves the Markov chains later too). **Exit time:** $\tau: \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$, $\tau = \min\{n \geq 0 : |X_n| \geq a\}$ (∞ if there is no such n); $m_i = \mathbb{E}_i[\tau]$.

Head-on

$\mathbb{P}_i(\tau = n)$ counts paths that stay strictly inside $(-a, a)$ for $n-1$ steps and exit at step n ; then $\sum_n n \mathbb{P}_i(\tau = n)$. The counts have no simple closed form.

First-step analysis

Condition on the first step, exactly as for the waiting time for HH (Conditional Expectation lecture, Application 2). After one step the walk is at $i \pm 1$ and, by independent increments, faces a fresh copy of the problem from there:

$$m_i = 1 + \frac{1}{2}m_{i+1} + \frac{1}{2}m_{i-1} \quad (-a < i < a), \quad m_{-a} = m_a = 0.$$

Take one step, pay one unit of time, and average the remaining expected times over the two places you might be.

- The unknowns $m_{-a+1}, \dots, m_{a-1}$ appear on both sides: it is a linear system of $2a-1$ equations, not a formula. This is the shape of every first-step argument in this lecture.

First-Step Analysis: The Solution

Theorem

$m_i = \mathbb{E}_i[\tau] = a^2 - i^2$ for $-a \leq i \leq a$; in particular $\mathbb{E}_0[\tau] = a^2$.

Proof

Try $m_i = c - i^2$: then $1 + \frac{1}{2}(c - (i+1)^2) + \frac{1}{2}(c - (i-1)^2) = 1 + c - i^2 - 1 = c - i^2$, so the equation holds for every c , and $m_{\pm a} = 0$ forces $c = a^2$. The system is $2a - 1$ linear equations in $2a - 1$ unknowns; it has a unique solution (uniqueness: if m, m' both solve it, $d = m - m'$ satisfies $d_i = \frac{1}{2}(d_{i+1} + d_{i-1})$ with $d_{\pm a} = 0$, so d is linear and vanishes). $\square$

The expected exit time is a quadratic in the starting point: reaching distance a costs a^2 steps on average.

$a = 3$

i	-3	-2	-1	0	1	2	3
m_i	0	5	8	9	8	5	0

From 0 the walk needs 9 steps on average to reach ± 3 ; from ± 2 , only 5.

This is the $\sqrt{n}$ scaling of slide 7 from the other side: distance $\sqrt{n}$ in n steps, or n steps for distance $\sqrt{n}$.

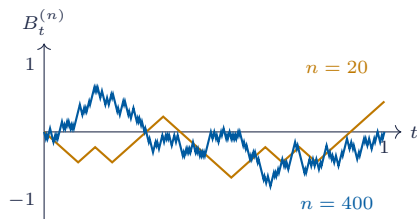
From the Walk to Brownian Motion: The $\sqrt{n}$ Scaling

A particle takes a fair ± 1 step every $\frac{1}{n}$ seconds, each of size $\frac{1}{\sqrt{n}}$. What does its path look like when n is large, and what is the probability that it is above 0 at time $t = 1$ and again at $t = 2$?

Rescaling

Position at time t is $B_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$. By slide 7, $\mathbb{E}[B_t^{(n)}] = 0$ and $\text{Var}(B_t^{(n)}) = \frac{\lfloor nt \rfloor}{n} \rightarrow t$; by the central limit theorem (CLT: for i.i.d. ξ_k with mean 0 and variance 1, $\mathbb{P}(S_m/\sqrt{m} \leq x) \rightarrow \Phi(x)$ for every x , with $\Phi: \mathbb{R} \rightarrow [0, 1]$ the $N(0, 1)$ cdf), $\mathbb{P}(B_t^{(n)} \leq x) \rightarrow \Phi(x/\sqrt{t})$: $B_t^{(n)} \rightarrow N(0, t)$ in distribution. Increments over disjoint time intervals are sums of disjoint blocks of steps, so they are independent, with variance equal to the length of the interval.

Dividing by $\sqrt{n}$ is the only scaling that keeps the spread finite and non-zero; the limit has Gaussian columns and independent increments.



The same coin sequence, 20 and 400 steps, each rescaled to fit $[0, 1]$ with steps of size $1/\sqrt{n}$. At time 1 both have variance 1.

The Invariance Principle

Theorem (invariance principle, Donsker 1951; informal statement)

Let $\xi_1, \xi_2, \dots$ be independent with $\mathbb{E}[\xi_k] = 0$ and $\text{Var}(\xi_k) = 1$, $S_n = \xi_1 + \dots + \xi_n$, and $B_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$. Then, as $n \rightarrow \infty$, the random path $t \mapsto B_t^{(n)}$ converges in distribution to a random continuous path $t \mapsto B_t$, and for all times $t_1 < \dots < t_k$ and sets $A_1, \dots, A_k$,

$$\mathbb{P}(B_{t_1}^{(n)} \in A_1, \dots, B_{t_k}^{(n)} \in A_k) \rightarrow \mathbb{P}(B_{t_1} \in A_1, \dots, B_{t_k} \in A_k).$$

The limit process B is the same for every step distribution with mean 0 and variance 1.

Seen from far enough away, every mean-zero random walk with unit-variance steps looks like the same random curve, and that curve is Brownian motion.

- Not proved here. “Invariance”: the limit does not depend on the step distribution, as in the central limit theorem, which is the one-column case $t_1 = 1$.
- The limit inherits from the walk: independent increments over disjoint intervals (sums over disjoint blocks of steps), $\text{Var}(\text{increment}) = \text{length}$, and a start at 0. Steps of size $1/\sqrt{n} \rightarrow 0$ deliver continuous paths. These are precisely the four properties in the definition on the next slide.

Brownian Motion: Definition

Definition

A **(standard) Brownian motion** is a process $B: [0, \infty) \times \Omega \rightarrow \mathbb{R}$, $(t, \omega) \mapsto B_t(\omega)$, such that

- (i) $B_0 = 0$;
- (ii) **independent increments:** for all $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $B_{t_1} - B_{t_0}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;
- (iii) **Gaussian increments:** for $0 \leq s < t$, $B_t - B_s \sim N(0, t - s)$, i.e. it has pdf
$$x \mapsto \frac{1}{\sqrt{2\pi(t-s)}} e^{-x^2/(2(t-s))};$$
- (iv) **continuous paths:** for every ω , the function $[0, \infty) \rightarrow \mathbb{R}$, $t \mapsto B_t(\omega)$, is continuous.

Over any time interval the process moves by an independent normal amount whose variance is the length of the interval, and it does so without jumping.

- Existence is a theorem (Wiener, 1923); we take it as given. The four properties determine all finite-dimensional distributions (slide 5), hence every probability about B .

What B Is: Reading the Table

The table of paths now has a continuum of columns ($t \geq 0$) and of rows (ω); the two readings are unchanged.

Column t : the random variable $B_t: \Omega \rightarrow \mathbb{R}$

$B_t = B_t - B_0 \sim N(0, t)$ by (i) and (iii). Its pdf is $f_{B_t}: \mathbb{R} \rightarrow [0, \infty)$, $f_{B_t}(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$, so $\mathbb{P}(a \leq B_t \leq b) = \int_a^b f_{B_t}(x) dx$.

Row ω : the path $[0, \infty) \rightarrow \mathbb{R}$, $t \mapsto B_t(\omega)$

A continuous function that starts at 0. Different rows are different functions; a typical one is as jagged as the $n = 400$ picture at every magnification. Row questions: the maximum up to time t , the first time the path reaches a , the time spent above 0.

Two arguments, two roles

In $B_t(\omega)$, t is the time (which column) and ω is the outcome (which row). Fix ω and vary t : a function of time. Fix t and vary ω : a random variable. An increment $B_t - B_s$ is a difference of two columns.

Independent increments is the whole content of the definition; every property below is a consequence of writing a value as a sum of increments.

What B Is Not

Brownian motion is **not**:

- a process with independent *values*: B_s and B_t are strongly dependent (next slide); it is the *increments* over disjoint intervals that are independent;
- stationary in value: $\text{Var}(B_t) = t$ grows; it is the increments whose distribution depends only on the interval length;
- smooth: no path has a derivative anywhere (slide 24);
- a single normal random variable: it is one normal per t , and jointly they form a Gaussian family.

Same structure, different time set and step distribution: B is the walk with the discreteness removed.

	walk S_n	Brownian B_t
time set T	$\{0, 1, 2, \dots\}$	$[0, \infty)$
state space	$\mathbb{Z}$	$\mathbb{R}$
start	$S_0 = 0$	$B_0 = 0$
increment over $(s, t]$	sum of $t - s$ steps	$N(0, t - s)$
Var of increment	$t - s$ (fair)	$t - s$
disjoint increments	independent	independent
path	moves by ± 1	continuous, rough

Basic Properties: Marginals and Covariance

Theorem

- (a) $B_t \sim N(0, t)$: $\mathbb{E}[B_t] = 0$, $\text{Var}(B_t) = t$.
- (b) $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t] = \min(s, t)$ for all $s, t \geq 0$.

Each column is a centred normal with variance t , and two columns overlap by the shorter of the two times.

Proof

(a) is (i) and (iii) with $s = 0$. (b) For $s < t$ write $B_t = B_s + (B_t - B_s)$. The increment is independent of B_s (property (ii) applied to $0 < s < t$) and has mean 0, so

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s (B_t - B_s)] = s + \mathbb{E}[B_s] \mathbb{E}[B_t - B_s] = s + 0 = \min(s, t).$$

Since $\mathbb{E}[B_s] = 0$, $\text{Cov}(B_s, B_t) = \mathbb{E}[B_s B_t]$. □

- The correlation of B_s and B_t is $\min(s, t)/\sqrt{st} = \sqrt{s/t}$ for $s < t$: B_1 and B_2 have correlation $1/\sqrt{2} \approx 0.71$; B_1 and B_{100} have correlation 0.1. Values are dependent, increments are not.
- Head-on, $\mathbb{E}[B_s B_t]$ is a two-dimensional integral against the joint pdf of the next slide. The one-line proof never writes it down.

Basic Properties: Joint Probability Density Function (pdf) of Several Columns

Theorem

For $0 = t_0 < t_1 < \dots < t_n$, the vector $(B_{t_1}, \dots, B_{t_n})$ has joint pdf $f: \mathbb{R}^n \rightarrow [0, \infty)$,

$$f(x_1, \dots, x_n) = \prod_{k=1}^n \frac{1}{\sqrt{2\pi(t_k - t_{k-1})}} \exp\left(-\frac{(x_k - x_{k-1})^2}{2(t_k - t_{k-1})}\right), \quad x_0 = 0.$$

Any finite set of columns has a density, and it is the product of the increment densities written in terms of the values.

Proof

The increments $D_k = B_{t_k} - B_{t_{k-1}}$, $k = 1, \dots, n$, are independent $N(0, t_k - t_{k-1})$ by (ii) and (iii), so $(D_1, \dots, D_n)$ has the product pdf $\prod_k \frac{1}{\sqrt{2\pi(t_k - t_{k-1})}} e^{-d_k^2/(2(t_k - t_{k-1}))}$. The values are

$B_{t_k} = D_1 + \dots + D_k$: a linear map from $(d_1, \dots, d_n)$ to $(x_1, \dots, x_n)$ with inverse $d_k = x_k - x_{k-1}$, of determinant 1. A determinant-one change of variables leaves a pdf unchanged apart from the substitution. □

Basic Properties: Scaling, Symmetry, Restart

Theorem

Let B be a Brownian motion. Each of the following is again a Brownian motion:

- (a) **Scaling:** $\left(\frac{1}{\sqrt{c}} B_{ct}\right)_{t \geq 0}$ for any fixed $c > 0$.
- (b) **Symmetry:** $(-B_t)_{t \geq 0}$.
- (c) **Restart:** $(B_{s+t} - B_s)_{t \geq 0}$ for any fixed $s \geq 0$, and it is independent of $(B_u)_{0 \leq u \leq s}$.

Brownian motion looks the same at every magnification, upside down, and from any moment onwards.

Proof

Check (i)–(iv). (a) $\frac{1}{\sqrt{c}} B_{c \cdot 0} = 0$; increments $\frac{1}{\sqrt{c}}(B_{ct} - B_{cs})$ over disjoint intervals are independent, normal with variance $\frac{1}{c}(ct - cs) = t - s$; paths are continuous. (b) $-N(0, \sigma^2)$ is $N(0, \sigma^2)$. (c) Increments of the new process are increments of B over intervals inside $[s, \infty)$, independent of increments over intervals inside $[0, s]$, and B_u for $u \leq s$ is such an increment. □

(a) with $c = n$ says $\frac{1}{\sqrt{n}} B_{nt}$ is Brownian: the picture on slide 13 in reverse; zooming into a Brownian path shows a Brownian path. (c) is the restart property of first-step analysis in continuous time: after any fixed time the process is a fresh copy, shifted to its current position.

Return to the Problem: $\mathbb{P}(B_2 > 0 \mid B_1 = 1)$

By the independent increment

$B_2 = B_1 + (B_2 - B_1)$, and $B_2 - B_1 \sim N(0, 1)$ is independent of B_1 . On the row $\{B_1 = 1\}$, $B_2 = 1 + (B_2 - B_1)$, so given $B_1 = 1$, $B_2 \sim N(1, 1)$ and

$$\mathbb{P}(B_2 > 0 \mid B_1 = 1) = \mathbb{P}(N(1, 1) > 0) = \mathbb{P}(N(0, 1) > -1) = \Phi(1) \approx 0.841,$$

with Φ the $N(0, 1)$ cdf of slide 13.

Head-on, for comparison

By slide 19 with $t_1 = 1$, $t_2 = 2$, (B_1, B_2) has joint pdf $f(x, y) = \frac{1}{2\pi} e^{-x^2/2} e^{-(y-x)^2/2}$, a bivariate normal with covariance matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Form $f_{B_2|B_1}(y \mid x) = f(x, y)/f_{B_1}(x) = \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}$ and integrate over $y > 0$ at $x = 1$. Same answer, after the same algebra, hidden.

Conditioning on B_1 fixes the first increment and leaves the second untouched; every conditional question about B is answered by splitting into increments.

- The same split gives the conditional distribution of B_t given $B_s = x$ for any $s < t$: $N(x, t - s)$. The past enters only through the present value.

Return to the Opening Problem: $\mathbb{P}(B_1 > 0, B_2 > 0)$

Let $X = B_1$ and $Z = B_2 - B_1$: independent, both $N(0, 1)$, and $B_2 = X + Z$. The event is $\{X > 0, X + Z > 0\}$.

By the independent increment and symmetry

Split according to the sign of Z :

$\{X > 0, X + Z > 0\} = \{X > 0, Z > 0\} \cup \{X > 0, Z < 0, X > -Z\}$. The first event has probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ by independence. For the second: on the row $\{X > 0, Z < 0\}$, which has probability $\frac{1}{4}$, the pair $(X, -Z)$ consists of two independent positive copies of the same distribution, so $X > -Z$ has conditional probability $\frac{1}{2}$ by symmetry. Hence

$$\mathbb{P}(B_1 > 0, B_2 > 0) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

Head-on, for comparison

Integrate the joint pdf $\frac{1}{2\pi} e^{-x^2/2} e^{-(y-x)^2/2}$ over the quadrant $x > 0, y > 0$. This is the orthant probability of a bivariate normal with correlation $\rho = 1/\sqrt{2}$, which by a polar-coordinate computation equals $\frac{1}{4} + \frac{\arcsin \rho}{2\pi} = \frac{1}{4} + \frac{\pi/4}{2\pi} = \frac{3}{8}$.

The particle that is above 0 at time 1 is above 0 at time 2 with probability $\frac{3/8}{1/2} = \frac{3}{4}$, not $\frac{1}{2}$: the values are dependent even though the increments are not.

The Reflection Principle for Brownian Motion

Theorem (reflection principle)

For $a > 0$ and $t > 0$: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a) = \mathbb{P}(|B_t| \geq a)$.

The maximum up to time t has the distribution of $|B_t|$: a path question is again a column question.

- Inherited from slide 8: the walk's middle term $\mathbb{P}(S_n = m)$ vanishes after the $\sqrt{n}$ scaling.

Example: $\mathbb{P}(\max_{s \leq 1} B_s \geq 1) = 2(1 - \Phi(1)) \approx 0.317$; head-on this would need the joint law of a continuum of columns.

Corollary (maximum and first hitting time)

Let $M_t = \max_{s \leq t} B_s$ and $T_a: \Omega \rightarrow [0, \infty]$, $T_a = \min\{s \geq 0 : B_s = a\}$. Then M_t has the distribution of $|B_t|$, and

$$\mathbb{P}(T_a \leq t) = \mathbb{P}(M_t \geq a) = 2(1 - \Phi(a/\sqrt{t})) \xrightarrow{t \rightarrow \infty} 1, \quad \text{yet} \quad \mathbb{E}[T_a] = \int_0^\infty \mathbb{P}(T_a > t) dt = \infty,$$

since $\mathbb{P}(T_a > t) = 2\Phi(a/\sqrt{t}) - 1 \approx a\sqrt{2/(\pi t)}$ for large t .

Every level is reached with probability one, but the mean time to reach it is infinite, as for the fair walk on slide 33.

Why the Paths Are Nowhere Differentiable

Theorem (the paths are rough)

With probability one, $t \mapsto B_t(\omega)$ is nowhere differentiable (Paley, Wiener, Zygmund, 1933; not proved here). For $t > 0$ and $\Delta_k = B_{kt/n} - B_{(k-1)t/n}$, $k = 1, \dots, n$: $\mathbb{E} \left[\sum_{k=1}^n \Delta_k^2 \right] = t$ and

$$\text{Var} \left(\sum_{k=1}^n \Delta_k^2 \right) = \frac{2t^2}{n} \rightarrow 0.$$

Why

By (iii), $\frac{B_{t+h} - B_t}{h} \sim N\left(0, \frac{1}{h}\right)$: the difference quotient has standard deviation $1/\sqrt{h} \rightarrow \infty$ as $h \rightarrow 0$, so it cannot converge. The Δ_k are independent $N(0, \frac{t}{n})$ with $\mathbb{E}[\Delta_k^2] = \frac{t}{n}$ and $\text{Var}(\Delta_k^2) = 2(\frac{t}{n})^2$; add.

The squared increments of a Brownian path over $[0, t]$ add up to t , which the squared increments of a differentiable function, each of order h^2 , never do.

- Inherited from the walk: the n steps of S_n have squared increments $(\pm 1)^2$ summing to exactly n , so the rescaled walk $B^{(n)}$ of slide 13 has $\sum (\Delta B^{(n)})^2 = n \cdot \frac{1}{n} = 1$ over $[0, 1]$, for every n ; the limit keeps this.
- The increments themselves do not add up: $\mathbb{E}|\Delta_k| = \sqrt{2t/(\pi n)}$ (the mean of $|N(0, \sigma^2)|$ is $\sigma\sqrt{2/\pi}$), so $\mathbb{E} \left[\sum_{k=1}^n |\Delta_k| \right] = \sqrt{2tn/\pi} \rightarrow \infty$: over any interval the path has infinite length.

Risk of Ruin: The Problem

A gambler with fortune $\mathcal{L}i$ bets $\mathcal{L}1$ per round, winning $\mathcal{L}1$ with probability p and losing it with probability $q = 1 - p$, rounds independent. She stops when her fortune reaches 0 (**ruin**) or a target $N > i$.

The opening question

Red at American roulette: $p = \frac{18}{38}$. Starting with $\mathcal{L}50$, aiming for $\mathcal{L}100$, one pound at a time. What is the probability of getting there, and how long does the evening last?

Head-on

The probability of reaching N is $\sum_n \mathbb{P}(\text{the game ends at round } n \text{ with fortune } N)$: paths of every length n that stay strictly between 0 and N until the last step. No column of the table answers this; the summands have no closed form, and there are infinitely many of them.

The question is about where a path exits, not about its value at any fixed time: a row question with no bound on the length of the row, asked alike of a gambler, an insurer and a queue.

Risk of Ruin: The Setting

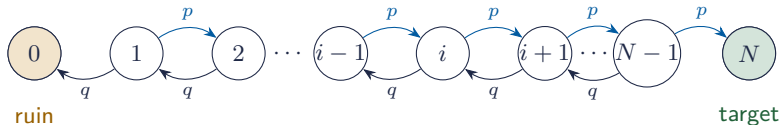
Definition

With $\xi_1, \xi_2, \dots$ the Bernoulli process of slide 6, the fortune is the walk started at i and stopped on exit from $\{1, \dots, N-1\}$:

$$X_n = i + S_n \text{ for } n \leq T, \quad T = \min\{n \geq 0 : X_n \in \{0, N\}\}, \quad h_i = \mathbb{P}_i(X_T = N).$$

The function $h: \{0, 1, \dots, N\} \rightarrow [0, 1]$ is the **probability of reaching the target** from fortune i ; $1 - h_i$ is the **probability of ruin**.

The gambler's fortune is a random walk between two walls, and h_i is the chance that the walk started at i touches the right wall first.



- T is a random variable $\Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$; that $T < \infty$ with probability one follows from the analysis below (the expected duration is finite).

First-Step Analysis: The Ruin Equation

Theorem

The function h satisfies

$$h_i = p h_{i+1} + q h_{i-1} \quad (0 < i < N), \quad h_0 = 0, \quad h_N = 1.$$

After one round the gambler has $i + 1$ or $i - 1$ and faces the same problem from there, so her chance is the average of the two neighbouring chances.

Proof

$h_0 = 0$ and $h_N = 1$: the game is already over. For $0 < i < N$, condition on the first round (law of total probability with the two rows $\{\xi_1 = +1\}$ and $\{\xi_1 = -1\}$):

$$h_i = \mathbb{P}_i(X_T = N \mid \xi_1 = +1) p + \mathbb{P}_i(X_T = N \mid \xi_1 = -1) q.$$

On the row $\{\xi_1 = +1\}$ the fortune is $i + 1$ after round one and, by independent increments, the remaining rounds $\xi_2, \xi_3, \dots$ are a fresh Bernoulli process: the conditional probability of reaching N is h_{i+1} . Likewise h_{i-1} on the other row. □

- The waiting-time argument of the Conditional Expectation lecture (Application 2) and of slide 11, now for a probability: $N - 1$ linear equations in $h_1, \dots, h_{N-1}$, each looking one step ahead.

The Fair Game: $h_i = i/N$

Theorem ($p = q = \frac{1}{2}$)

$$h_i = \frac{i}{N} \text{ for } i = 0, \dots, N. \quad \text{Probability of ruin: } 1 - \frac{i}{N}.$$

In a fair game your chance of turning i into N is exactly the fraction of the target you already hold.

Proof

The equation $h_i = \frac{1}{2}h_{i+1} + \frac{1}{2}h_{i-1}$ says $h_{i+1} - h_i = h_i - h_{i-1}$: consecutive differences are equal, so h is linear in i , $h_i = A + Bi$. Then $h_0 = 0$ gives $A = 0$ and $h_N = 1$ gives $B = \frac{1}{N}$. Uniqueness: if h, h' both solve the system, $d = h - h'$ is linear with $d_0 = d_N = 0$, hence $d = 0$. $\square$

A second look: the fair game preserves expected fortune

Each round has mean gain 0, so $\mathbb{E}[X_T] = X_0 = i$ (this needs an argument for the random time T , which we omit). But $X_T \in \{0, N\}$, so $\mathbb{E}[X_T] = N h_i$. Hence $h_i = i/N$ without solving any equation.

- £50 to £100 at a fair game: $\frac{1}{2}$. £3 to £10: $\frac{3}{10}$. £1 to £10: $\frac{1}{10}$, and ruin $\frac{9}{10}$.

The Unfair Game: $h_i = \frac{r^i - 1}{r^N - 1}$ with $r = q/p$

Theorem ($p \neq q$)

With $r = q/p \neq 1$: $h_i = \frac{r^i - 1}{r^N - 1}$, probability of ruin $1 - h_i = \frac{r^N - r^i}{r^N - 1}$.

The chances now grow geometrically in i rather than linearly, with ratio q/p , so a small edge compounds over every pound between you and the target.

Proof

Try $h_i = r^i$ in $h_i = ph_{i+1} + qh_{i-1}$: the right side is $r^{i-1}(pr^2 + q)$, and $pr^2 + q = \frac{q^2}{p} + q = \frac{q(q+p)}{p} = r$, so it equals r^i . Constants also solve the equation. Hence $h_i = A + Br^i$ solves it for every A, B , and since a solution is fixed by h_0, h_1 (the equation determines h_{i+1} from h_i, h_{i-1}), these are all the solutions. Now $h_0 = 0$ gives $A = -B$ and $h_N = 1$ gives $B(r^N - 1) = 1$. $\square$

$p = 0.4, r = 1.5$

From £5 to £10: $h_5 = \frac{1.5^5 - 1}{1.5^{10} - 1} = \frac{32}{275} \approx 0.116$, against $\frac{1}{2}$ in the fair game. From £3 to £10: $h_3 \approx 0.042$, against 0.3.

Expected Duration of the Game

Let $D_i = \mathbb{E}_i[T]$, so $D: \{0, \dots, N\} \rightarrow [0, \infty)$.

Theorem

$D_i = 1 + p D_{i+1} + q D_{i-1}$ for $0 < i < N$, $D_0 = D_N = 0$, and the solution is

$$D_i = i(N-i) \quad (p = q), \quad D_i = \frac{i}{q-p} - \frac{N}{q-p} \cdot \frac{1-r^i}{1-r^N} \quad (p \neq q, r = q/p).$$

The fair game from the middle lasts about $N^2/4$ rounds, the square of the distance to the boundary, as on slide 12.

Proof

The equation is first-step analysis with one round paid before moving to $i \pm 1$, as on slide 11. For $p = q$ it is the equation of slide 12 on $\{0, \dots, N\}$; $D_i = i(N-i)$ satisfies it since $\frac{1}{2}((i+1)(N-i-1) + (i-1)(N-i+1)) = i(N-i) - 1$, and it vanishes at 0 and N . For $p \neq q$: $D_i = \frac{i}{q-p}$ is one solution of the equation with the $+1$ ($\frac{p(i+1)+q(i-1)}{q-p} = \frac{i-(q-p)}{q-p} = \frac{i}{q-p} - 1$); add the general solution $A + Br^i$ of the homogeneous equation and fit $D_0 = D_N = 0$. Uniqueness as before. $\square$

- Fair, £50 to £100: 2500 rounds on average. Roulette ($p = \frac{18}{38}$), same stakes: about 940 rounds. Losing is not only likelier, it is quicker.

The Casino's Edge in Numbers, and Why Doubling Fails

$i \rightarrow N$ (stake £1)	p	$\mathbb{P}(\text{target})$	$\mathbb{E}[T]$
50 $\rightarrow$ 100	0.5	0.500	2500
50 $\rightarrow$ 100	0.49	0.119	1904
50 $\rightarrow$ 100	18/38	0.005	940
10 $\rightarrow$ 20	18/38	0.259	92
5 $\rightarrow$ 10	18/38	0.371	24
1 $\rightarrow$ 2	18/38	0.474	1

At a 1% disadvantage, doubling £50 one pound at a time succeeds one time in eight; the last three rows are the same goal with £5, £10, £50 stakes, and fewer rounds means fewer chances for the edge to compound.

Doubling (the “martingale” system)

Bet 1; after each loss double the stake; after a win pocket the net £1 and restart. With capital $2^k - 1$ you can absorb $k - 1$ losses; k losses in a row ruin you. Per cycle: gain +1 with probability $1 - q^k$, lose $2^k - 1$ with probability q^k :

$$\mathbb{E}[\text{gain}] = (1 - q^k) - q^k(2^k - 1) = 1 - (2q)^k.$$

This is 0 if $q = \frac{1}{2}$ and negative if $q > \frac{1}{2}$: the strategy changes the *shape* of the outcome (many small wins, one rare catastrophe), not its mean.

- Roulette, $k = 7$ (capital £127): win £1 with probability 0.989, lose £127 with probability 0.011; mean $-\text{£}0.43$ per cycle.

The Infinite Target: Playing Against a Casino That Cannot Be Broken

Let $N \rightarrow \infty$ with i fixed: the gambler plays on until ruin or forever.

Theorem

Let $\rho_i = \mathbb{P}_i(\text{ruin, ever}) = \lim_{N \rightarrow \infty} (1 - h_i)$. Then

$$\rho_i = 1 \quad (p \leq q), \quad \rho_i = \left(\frac{q}{p}\right)^i \quad (p > q).$$

In the fair case, moreover, $\mathbb{E}[\text{time to ruin}] = \lim_N i(N - i) = \infty$.

Against unlimited capital, a fair or unfavourable game ends in ruin with certainty; only a favourable game leaves a positive chance of playing forever, and even then it shrinks geometrically in the initial deficit.

Proof

Fair: $1 - h_i = 1 - \frac{i}{N} \rightarrow 1$. Unfair with $r = q/p$: $1 - h_i = \frac{r^N - r^i}{r^N - 1}$. If $r > 1$ ($p < q$), divide by r^N :
 $\rightarrow \frac{1 - 0}{1 - 0} = 1$. If $r < 1$ ($p > q$), $r^N \rightarrow 0$ and $1 - h_i \rightarrow \frac{0 - r^i}{0 - 1} = r^i$. □

The Infinite Target: Three Regimes

game	$\mathbb{P}(\text{ruin})$ from i	$\mathbb{E}[\text{time to ruin}]$	the walk $i + S_n$
unfavourable, $p < q$	1	$\frac{i}{q-p} < \infty$	drifts down; hits 0 soon
fair, $p = q$	1	∞	hits 0 eventually, but slowly
favourable, $p > q$	$(q/p)^i < 1$	∞ (may never be ruined)	drifts up; may never hit 0

- $p = 0.6$, £5 in hand: ruin with probability $(\frac{2}{3})^5 \approx 0.13$; otherwise the fortune drifts to infinity.
- Roulette, £50 in hand, no target: ruin certain, after $\frac{50}{2/38} = 950$ spins on average.
- Fair: ruined with probability one, yet no finite number of rounds is a fair estimate of when. This is the statement that the fair walk returns to any level with probability one but has infinite expected return time; it reappears as *null recurrence* on slide 60.
- ρ_i for $p > q$ is the probability that a walk with upward drift ever falls by i below its start: the **hitting probability** of level $-i$. Hitting probabilities are the subject of slides 61–64, where the ruin problem returns as a special case.

One formula, three regimes: certain ruin fast, certain ruin slowly, or a geometric chance of survival.

Return to the Opening Problem

Red at roulette, $p = \frac{18}{38}$, $r = \frac{20}{18} = \frac{10}{9}$. From £50, one pound per spin:

target N	$h_{50} = \frac{r^{50} - 1}{r^N - 1}$	ruin	$\mathbb{E}[T]$ (spins)
100 (double)	0.005	0.995	940
60 (win £10)	0.348	0.652	554
55 (win £5)	0.589	0.411	334
51 (win £1)	0.900	0.100	78

- Doubling £50 at £1 a spin: one chance in 195, and the evening lasts about 940 spins. Staking the whole £50 on one spin: $\frac{18}{38} \approx 0.474$.
- Even “win £1 and leave” fails one time in ten, and the failures cost £50 each: mean gain $0.8995 \cdot 1 - 0.1005 \cdot 50 \approx -£4.12$, which is $(p - q) \mathbb{E}[T] = -\frac{2}{38} \cdot 78.4$. No staking plan makes the mean positive: every plan is a sum of bets each with mean $-\frac{2}{38}$ per pound.

Two lines of first-step analysis produce every number in this table; no path was ever enumerated.

Markov Chains: The Problem

A town's weather is Sunny, Cloudy or Rainy each day, and tomorrow's weather depends on today's according to the table below. It is sunny today.

1. What is the probability that it is rainy a week from now?
2. In the long run, what fraction of days are sunny?
3. How many days until the next rainy day, on average?

Head-on

Question 1 sums over $3^6 = 729$ weather sequences for the six intermediate days. Question 2 needs a limit over sequences of every length. Question 3 is a waiting time whose distribution is a mess.

Each question is about a process whose next value depends only on its current value; that single structural fact will answer all three with matrix arithmetic.

today \ tomorrow	S	C	R
S	0.7	0.2	0.1
C	0.3	0.4	0.3
R	0.2	0.4	0.4

Row S reads: after a sunny day, tomorrow is sunny with probability 0.7, cloudy 0.2, rainy 0.1. Each row sums to 1.

Markov Chain: Definition

Definition

Let S be a finite or countable set (the **state space**). A process $(X_n)_{n \geq 0}$ with $X_n: \Omega \rightarrow S$ is a **Markov chain** if for all $n \geq 0$ and all states $i_0, \dots, i_{n-1}, i, j \in S$ with $\mathbb{P}(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) > 0$,

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad (\text{Markov property}),$$

and it is **homogeneous** if the right side does not depend on n . Then the **transition probabilities** are the function

$$p: S \times S \rightarrow [0, 1], \quad p(i, j) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Given the whole past, the next state depends only on the present state, and the rule for moving on is the same at every time.

- Two arguments, two roles: i is the *current* state (the conditioning event $\{X_n = i\}$, the row); j is the *next* state (the column). $p(i, j)$ is the probability of the step $i \rightarrow j$.
- Random walk: $S = \mathbb{Z}$, $p(i, i+1) = p$, $p(i, i-1) = q$, all other $p(i, j) = 0$. The Markov property holds because $S_{n+1} = S_n + \xi_{n+1}$ with ξ_{n+1} independent of the past. Gambler's ruin: the same on $S = \{0, \dots, N\}$ with $p(0, 0) = p(N, N) = 1$.
- All chains below are homogeneous.

The Transition Matrix: A Table Whose Rows Are Conditional Distributions

Definition

The **transition matrix** is p written as a table,
 $\mathbf{P} = (p(i, j))_{i, j \in S}$: row i , column j . It is a **stochastic matrix**:

$$p(i, j) \geq 0 \quad \text{and} \quad \sum_{j \in S} p(i, j) = 1 \quad \text{for every } i \in S.$$

The **initial distribution** is the row vector $\lambda: S \rightarrow [0, 1]$,
 $\lambda_i = \mathbb{P}(X_0 = i)$, $\sum_i \lambda_i = 1$.

Row i of $\mathbf{P}$ is the conditional pmf of X_{n+1} given $X_n = i$: the “restrict to the row” picture of the two-dice table.

- Row i , as a function $j \mapsto p(i, j)$, is $p_{X_{n+1}|X_n}(\cdot | i)$: non-negative, sums to 1. Columns need not sum to anything.
- $\mathbb{E}[f(X_{n+1}) | X_n = i] = \sum_j p(i, j)f(j)$: the row average of f (compare $\mathbb{E}[X | Y = y]$, lecture 5).

		next state j		
		S	C	R
current state i	S	0.7	0.2	0.1
	C	0.3	0.4	0.3
	R	0.2	0.4	0.4

The weather chain. The shaded row is $p_{X_{n+1}|X_n}(\cdot | C)$: the distribution of tomorrow given a cloudy today.

Paths Have Product Probabilities

Theorem

For a Markov chain with initial distribution λ and transition matrix $\mathbf{P}$, and any states $i_0, i_1, \dots, i_n$,

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = \lambda_{i_0} p(i_0, i_1) p(i_1, i_2) \cdots p(i_{n-1}, i_n).$$

Conversely, a process whose joint pmfs have this form is a Markov chain with transition matrix $\mathbf{P}$.

The probability of a row of the table of paths is the start probability times the product of the step probabilities along the row; λ and $\mathbf{P}$ together determine every finite-dimensional distribution.

Proof

Multiplication rule:

$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_0 = i_0) \prod_{k=1}^n \mathbb{P}(X_k = i_k \mid X_0 = i_0, \dots, X_{k-1} = i_{k-1})$, and each factor is $p(i_{k-1}, i_k)$ by the Markov property. For the converse, divide two such products to get the conditional probability. □

Weather chain (slide 37), sunny today

$$\mathbb{P}(X_1 = \text{C}, X_2 = \text{R} \mid X_0 = \text{S}) = p(\text{S}, \text{C}) p(\text{C}, \text{R}) = 0.2 \cdot 0.3 = 0.06;$$

$$\mathbb{P}(X_1 = \text{R}, X_2 = \text{R}, X_3 = \text{S} \mid X_0 = \text{S}) = 0.1 \cdot 0.4 \cdot 0.2 = 0.008.$$

What the Markov Property Is Not

- **Not independence.** X_{n+1} depends on X_n , often strongly; it is the *earlier* past that adds nothing once X_n is known.
- **Not “the past is irrelevant”.** $\mathbb{P}(X_{n+1} = j \mid X_{n-1} = k) \neq \mathbb{P}(X_{n+1} = j)$ in general; the past is irrelevant only *given the present*.
- **Not every process.** Toss a coin and let X_n be the number of heads among tosses $n - 1$ and n . Given $X_n = 1$, the next value depends on whether the recent toss was the head:
 $\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 0) = \frac{1}{2}$ but $\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 2) = 0$. Not Markov; the pair of the last two tosses is.
- **Not a property of the phenomenon but of what is recorded.** A process that is not Markov often becomes Markov when the state is enlarged to carry the relevant part of the past (last two tosses; position *and* velocity; today's weather *and* yesterday's).

Test for the Markov property

Check that $\mathbb{P}(X_{n+1} = j \mid X_n = i, \text{ anything about } X_0, \dots, X_{n-1})$ does not change with the “anything”. In models built as $X_{n+1} = F(X_n, \xi_{n+1})$ with ξ_{n+1} independent of $X_0, \dots, X_n$, it holds automatically: the walk, the urn, the graph walks.

Markov is the statement that the present state is a sufficient summary of the past for predicting the future.

A Gallery of Chains (1)

Vowels and consonants

Model a text as a sequence of V and C. After a vowel, the next letter is a vowel with probability 0.2; after a consonant, with probability 0.5.

$$S = \{V, C\}, \quad \mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}.$$

A two-state chain; every question about it is a 2×2 computation. The Entropy lecture uses this chain for its entropy-rate example.

Gambler's ruin

$S = \{0, \dots, N\}$, $p(i, i+1) = p$, $p(i, i-1) = q$ for $0 < i < N$, and $p(0,0) = p(N,N) = 1$: the diagram of slide 26. States 0 and N are **absorbing**: once entered, never left.

The Ehrenfest urn (gas between two chambers)

N numbered balls in two urns. Each step, pick a ball uniformly at random and move it to the other urn. X_n is the number of balls in urn 1: $S = \{0, \dots, N\}$,

$$p(i, i-1) = \frac{i}{N}, \quad p(i, i+1) = \frac{N-i}{N}.$$

The more balls in an urn, the likelier the next move is out of it.

$$N = 4: \quad \mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

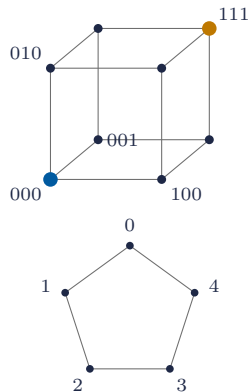
A Gallery of Chains (2): Walks on Graphs

Random walk on a graph

Vertices S ; from vertex i move to one of its $d(i)$ neighbours uniformly: $p(i, j) = 1/d(i)$ if j is a neighbour of i , else 0.

- **Cycle** of length L : $S = \{0, \dots, L-1\}$, $p(i, i \pm 1 \bmod L) = \frac{1}{2}$. The random walk with the ends joined.
- **Cube**: $S = \{0, 1\}^3$, the 8 corners; neighbours differ in one coordinate, so $p(i, j) = \frac{1}{3}$ for each of the 3 neighbours. A move flips one coordinate.
- **The web**: pages as vertices, links as directed edges, $p(i, j) = 1/(\text{number of links out of } i)$ for each page j that i links to. A **random surfer**. Slides 52–53.

On a graph, the transition matrix is the adjacency table with each row rescaled to sum to 1.



Left: the cube; from 000, the three neighbours 100, 010, 001. Right: the cycle $L = 5$.

n -Step Transition Probabilities

As on slide 11, $\mathbb{P}_i(\cdot) = \mathbb{P}(\cdot \mid X_0 = i)$ and $\mathbb{E}_i[\cdot] = \mathbb{E}[\cdot \mid X_0 = i]$ refer to the chain started at i (the case $\lambda = \text{point mass at } i$).

Definition

The **n -step transition probabilities** are the function $p^{(n)}: S \times S \rightarrow [0, 1]$,

$$p^{(n)}(i, j) = \mathbb{P}_i(X_n = j) = \mathbb{P}(X_{m+n} = j \mid X_m = i) \quad (\text{any } m, \text{ by homogeneity}),$$

with $p^{(1)} = p$ and $p^{(0)}(i, j) = \mathbf{1}\{i = j\}$. As before, i is the current state (row) and j the state n steps later (column).

The same table as $\mathbf{P}$, but for a jump of n steps: row i is the conditional distribution of X_{m+n} given $X_m = i$.

Weather: $p^{(2)}(S, R)$ by hand

Sum over tomorrow's weather k :

$p^{(2)}(S, R) = \sum_k p(S, k) p(k, R) = 0.7 \cdot 0.1 + 0.2 \cdot 0.3 + 0.1 \cdot 0.4 = 0.17$. Three two-day sequences, SSR, SCR, SRR, each a product from slide 38.

- That sum is the (S, R) entry of the matrix product $\mathbf{P}\mathbf{P}$. The next slide says this is general.

Chapman–Kolmogorov: n Steps Is the n -th Power

Theorem (Chapman–Kolmogorov)

For all $m, n \geq 0$ and $i, j \in S$: $p^{(m+n)}(i, j) = \sum_{k \in S} p^{(m)}(i, k) p^{(n)}(k, j)$. Equivalently, writing

$\mathbf{P}^{(n)} = (p^{(n)}(i, j))_{i, j}$ for the table of n -step probabilities: $\mathbf{P}^{(n)} = \mathbf{P}^n$, the n -th matrix power.

To go from i to j in $m + n$ steps, be somewhere at time m and go on from there: condition on the intermediate state, and the sum is exactly a matrix product.

Proof

Total probability with the rows $\{X_m = k\}$, $k \in S$:

$$\mathbb{P}_i(X_{m+n} = j) = \sum_k \mathbb{P}_i(X_m = k) \mathbb{P}_i(X_{m+n} = j \mid X_m = k) = \sum_k p^{(m)}(i, k) p^{(n)}(k, j),$$

the second factor by the Markov property and homogeneity (the past before time m , including $X_0 = i$, drops out). With $m = 1$ this is $\mathbf{P}^{(n+1)} = \mathbf{P} \mathbf{P}^{(n)}$, so by induction $\mathbf{P}^{(n)} = \mathbf{P}^n$. $\square$

- First-step analysis is the case $m = 1$; “last-step” is $n = 1$; a computer uses repeated squaring. Rows of $\mathbf{P}^n$ are again conditional distributions: $\mathbf{P}^n$ is stochastic.

The Distribution at Time n : Row Vector Times $\mathbf{P}^n$

Theorem

If X_0 has distribution λ (a row vector indexed by S), then X_n has distribution $\lambda \mathbf{P}^n$:

$$\mathbb{P}(X_n = j) = \sum_{i \in S} \lambda_i p^{(n)}(i, j) = (\lambda \mathbf{P}^n)_j.$$

Start with the row vector of start probabilities and multiply by $\mathbf{P}$ once per step; each multiplication averages the rows of $\mathbf{P}$ with the current weights.

Proof

Total probability over the rows $\{X_0 = i\}$: $\mathbb{P}(X_n = j) = \sum_i \mathbb{P}(X_0 = i) \mathbb{P}(X_n = j \mid X_0 = i)$. □

- Every finite-dimensional distribution follows: $\mathbb{P}(X_m = i, X_{m+n} = j) = (\lambda \mathbf{P}^m)_i p^{(n)}(i, j)$, and so on along any sequence of times.
- Two objects, two shapes: $\lambda \mathbf{P}^n$ is a row vector (a distribution on S); $\mathbf{P}^n$ is a matrix (a table of conditional distributions). Multiplying a distribution by $\mathbf{P}$ moves it one step forward in time.

Return to Question 1: Rainy a Week From a Sunny Day

$\lambda = (1, 0, 0)$, so the distribution of X_7 is row S of $\mathbf{P}^7$.

$$\mathbf{P}^2 = \begin{pmatrix} 0.57 & 0.26 & 0.17 \\ 0.39 & 0.34 & 0.27 \\ 0.34 & 0.36 & 0.30 \end{pmatrix}, \quad \mathbf{P}^3 = \begin{pmatrix} 0.511 & 0.286 & 0.203 \\ 0.429 & 0.322 & 0.249 \\ 0.406 & 0.332 & 0.262 \end{pmatrix}, \quad \mathbf{P}^7 = \begin{pmatrix} 0.4637 & 0.3068 & 0.2296 \\ 0.4601 & 0.3083 & 0.2316 \\ 0.4591 & 0.3087 & 0.2321 \end{pmatrix}.$$

Answer

$\mathbb{P}_S(X_7 = R) = p^{(7)}(S, R) \approx 0.230$. The head-on sum over 729 six-day sequences has been organised into six matrix multiplications.

- Read the three rows of $\mathbf{P}^7$: they are almost equal. A week on, the weather barely remembers today. The common row, $(0.4615, 0.3077, 0.2308)$ to four places from $\mathbf{P}^{20}$, is the subject of the next section.
- Read the columns of $\mathbf{P}^2$ versus $\mathbf{P}$: after a rainy day, the chance of sun is 0.2 tomorrow and 0.34 the day after; the rows are mixing towards each other.

Everything about the future of the chain is in the powers of one small table.

Powers of a 2×2 Chain: The Rows Converge

Vowel/consonant chain

$$\mathbf{P} = \begin{pmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{pmatrix}, \quad \mathbf{P}^2 = \begin{pmatrix} 0.44 & 0.56 \\ 0.35 & 0.65 \end{pmatrix}, \quad \mathbf{P}^3 = \begin{pmatrix} 0.368 & 0.632 \\ 0.395 & 0.605 \end{pmatrix}, \quad \mathbf{P}^{10} = \begin{pmatrix} 0.3846 & 0.6154 \\ 0.3846 & 0.6154 \end{pmatrix}.$$

Both rows approach $(\frac{5}{13}, \frac{8}{13}) = (0.384615\dots, 0.615385\dots)$: ten letters on, the chance of a vowel no longer depends on whether we started from a vowel or a consonant.

Exact formula

$\mathbf{P}$ has eigenvalues 1 and $0.2 + 0.5 - 1 = -0.3$ (trace minus 1), and

$$p^{(n)}(\mathbf{V}, \mathbf{V}) = \frac{5}{13} + \frac{8}{13}(-0.3)^n, \quad p^{(n)}(\mathbf{C}, \mathbf{V}) = \frac{5}{13} - \frac{5}{13}(-0.3)^n.$$

Check: $n = 1$ gives $\frac{5}{13} - \frac{2.4}{13} = 0.2$ and $\frac{5}{13} + \frac{1.5}{13} = 0.5$. The distance to the limit shrinks by a factor 0.3 per step and alternates in sign.

The rows of $\mathbf{P}^n$ forgetting their index is the phenomenon the rest of the lecture explains: what the limit row is, when it exists, and how fast it is reached.

Stationary Distribution: Definition

Definition

A **stationary distribution** of a chain with transition matrix $\mathbf{P}$ on S is a function $\pi: S \rightarrow [0, 1]$ with $\sum_{i \in S} \pi_i = 1$ and

$$\pi \mathbf{P} = \pi, \quad \text{i.e.} \quad \sum_{i \in S} \pi_i p(i, j) = \pi_j \quad \text{for every } j \in S.$$

A distribution over states that the chain does not change: start X_0 from π and every X_n has distribution π .

Theorem

If $X_0 \sim \pi$ and π is stationary, then $X_n \sim \pi$ for every n , and (X_n, X_{n+1}) has the same joint distribution for every n .

Proof

$\pi \mathbf{P}^n = (\pi \mathbf{P}) \mathbf{P}^{n-1} = \pi \mathbf{P}^{n-1} = \dots = \pi$, then the previous slide. $\mathbb{P}(X_n = i, X_{n+1} = j) = \pi_i p(i, j)$ does not involve n . □

Stationary Distribution: Two Readings and Existence

Balance

$\sum_i \pi_i p(i, j)$ is the probability mass flowing into j in one step from the distribution π ; stationarity says it equals the mass π_j already there, for every j . Mass leaves j at rate $\pi_j \sum_k p(j, k) = \pi_j$, so inflow equals outflow at every state.

Left eigenvector

$\pi \mathbf{P} = \pi$ says π is a left eigenvector of $\mathbf{P}$ for the eigenvalue 1, normalised to sum to 1. Every stochastic matrix has the eigenvalue 1, because the column vector $\mathbf{1}$ of ones satisfies $\mathbf{P}\mathbf{1} = \mathbf{1}$ (row sums), and left and right eigenvalues coincide.

Theorem (existence)

Every Markov chain on a finite state space has at least one stationary distribution.

A left eigenvector for the eigenvalue 1 exists by the argument above; that it can be chosen with non-negative entries is the content of the theorem.

- Not proved here (Perron–Frobenius, or a fixed-point argument). Uniqueness can fail: the gambler's ruin chain has infinitely many. On an infinite S existence can fail: the random walk on $\mathbb{Z}$ has none (any solution of $\pi \mathbf{P} = \pi$ is constant, and cannot sum to 1).

Computing π : A Linear System

$\pi \mathbf{P} = \pi$ is $|S|$ linear equations in $|S|$ unknowns, but they are dependent (they sum to the identity $\sum_j \pi_j = \sum_i \pi_i$); drop one and add $\sum_i \pi_i = 1$.

Weather, question 2: the long-run fraction of sunny days

Unknowns π_S, π_C, π_R . Columns S and C of $\pi \mathbf{P} = \pi$, and normalisation:

$$0.7\pi_S + 0.3\pi_C + 0.2\pi_R = \pi_S, \quad 0.2\pi_S + 0.4\pi_C + 0.4\pi_R = \pi_C, \quad \pi_S + \pi_C + \pi_R = 1.$$

The first gives $3\pi_S = 3\pi_C + 2\pi_R$, the second $6\pi_C = 2\pi_S + 4\pi_R$; solving,

$$\pi = \left(\frac{6}{13}, \frac{4}{13}, \frac{3}{13} \right) \approx (0.4615, 0.3077, 0.2308),$$

which is exactly the common row of $\mathbf{P}^{20}$ on slide 45. About 46% of days are sunny in the long run (the precise sense of “long run” is slide 72).

Vowel/consonant

$0.2\pi_V + 0.5\pi_C = \pi_V$, i.e. $0.8\pi_V = 0.5\pi_C$, with $\pi_V + \pi_C = 1$: $\pi = \left(\frac{5}{13}, \frac{8}{13} \right)$, the limit row of slide 46 and the weights in the Entropy lecture’s entropy rate.

Detailed Balance: A Shortcut for Reversible Chains

Theorem

If $\pi: S \rightarrow [0, 1]$ sums to 1 and satisfies the **detailed balance** equations

$$\pi_i p(i, j) = \pi_j p(j, i) \quad \text{for all } i, j \in S,$$

then π is stationary. Such a chain is called **reversible** (with respect to π).

If the probability flow along every single edge balances in both directions, then in particular the total flow into each state balances.

Proof

$$\sum_i \pi_i p(i, j) = \sum_i \pi_j p(j, i) = \pi_j \sum_i p(j, i) = \pi_j. \quad \square$$

- Stronger than stationarity ($|S|^2$ equations), but each involves only two unknowns, so it is solved by walking along edges; if it has no solution, π must be found from $\pi \mathbf{P} = \pi$.
- Random walk on a graph: $\pi_i = d(i) / \sum_k d(k)$ satisfies $\pi_i p(i, j) = \frac{1}{\sum_k d(k)} = \pi_j p(j, i)$ on every edge. Cube and cycle: uniform.
- Reversible means: started from π , the chain run backwards has the same law, since $\mathbb{P}(X_n = i, X_{n+1} = j) = \pi_i p(i, j) = \pi_j p(j, i) = \mathbb{P}(X_n = j, X_{n+1} = i)$.

Detailed Balance at Work: The Ehrenfest Urn

The Ehrenfest urn: π is binomial

Only neighbours communicate, so detailed balance is one equation per edge: $\pi_i \frac{N-i}{N} = \pi_{i+1} \frac{i+1}{N}$, i.e. $\pi_{i+1} = \pi_i \frac{N-i}{i+1}$. Iterating from π_0 gives $\pi_i = \pi_0 \binom{N}{i}$, and $\sum_i \pi_i = 1$ forces $\pi_0 = 2^{-N}$:

$$\pi_i = \binom{N}{i} 2^{-N} \quad (\text{each ball in urn 1 independently with probability } \tfrac{1}{2}).$$

$N = 4$: $\pi = \frac{1}{16}(1, 4, 6, 4, 1)$. Check one edge: $\pi_1 p(1, 2) = \frac{4}{16} \cdot \frac{3}{4} = \frac{3}{16} = \frac{6}{16} \cdot \frac{1}{2} = \pi_2 p(2, 1)$.

The long-run picture of the urn is N independent fair coins, one per ball, saying which urn the ball is in.

- Head-on, $\pi \mathbf{P} = \pi$ is a 5×5 system; for $N = 100$ it is 101×101 , while detailed balance is still one line per edge.
- $\pi_0 = 2^{-N}$, so by slide 59 the mean time between two moments when urn 1 is empty is 2^N steps: 16 for $N = 4$, about $1.3 \cdot 10^{30}$ for $N = 100$; meanwhile π is concentrated within a few $\sqrt{N}$ of $N/2$.

PageRank: The Stationary Distribution of a Random Surfer

Definition

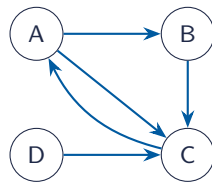
n web pages, links as directed edges. A **random surfer** moves as follows: with probability d (the damping factor, $d = 0.85$) follow a uniformly chosen link out of the current page; with probability $1 - d$ jump to a uniformly random page. Its transition matrix is

$$\mathbf{P} = d\mathbf{H} + \frac{1-d}{n}\mathbf{J}, \quad \mathbf{H}_{ij} = \frac{\mathbf{1}\{i \rightarrow j\}}{\#\{\text{links out of } i\}}, \quad \mathbf{J}_{ij} = 1.$$

The **PageRank** of page j is π_j , where $\pi: S \rightarrow [0, 1]$ is the stationary distribution of $\mathbf{P}$.

*A page is important if important pages link to it;
 $\pi_j = \sum_i \pi_i p(i, j)$ is that sentence as an equation.*

- Why the jump: without it, a page with no out-links (or a cluster with no exits) traps the surfer, and the walk on $\mathbf{H}$ alone may fail to have a unique π or to converge (slides 54–69). With $d < 1$ every entry of $\mathbf{P}$ is positive, and everything below applies.



Four pages. Links $A \rightarrow B, C$; $B \rightarrow C$; $C \rightarrow A$; $D \rightarrow C$. $\mathbf{P}$ adds a 0.0375 chance of jumping to each page from each page.

In $\mathbf{H}_{ij}$, i is the page the surfer is on and j the page linked to; row sums are 1 (for a page with no out-links put $\mathbf{H}_{ij} = 1/n$ for all j).

PageRank: The Four-Page Web

$$\mathbf{H} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P} = 0.85 \mathbf{H} + 0.0375 \mathbf{J} = \begin{pmatrix} 0.0375 & 0.4625 & 0.4625 & 0.0375 \\ 0.0375 & 0.0375 & 0.8875 & 0.0375 \\ 0.8875 & 0.0375 & 0.0375 & 0.0375 \\ 0.0375 & 0.0375 & 0.8875 & 0.0375 \end{pmatrix}.$$

Solving $\pi \mathbf{P} = \pi$, $\sum_j \pi_j = 1$

$$\pi \approx (0.373, 0.196, 0.394, 0.038) \quad \text{for (A, B, C, D),} \quad \text{exactly } \pi_D = \frac{3}{80} = 0.0375.$$

C ranks first: three pages link to it, one of them (A) itself highly ranked. A is second with a single in-link, because that link comes from the top page and is C's only link. D, with no in-links, gets exactly the jump mass $\frac{1-d}{n}$.

- Head-on, “importance” is circular: A is important because C is, and C because A is. The stationary equation resolves the circularity as a fixed point; the powers $\mathbf{P}^n$ compute it by iteration ($\mathbf{P}^{30}$ has all rows equal to π to four places).
- Without damping ($d = 1$), $\pi = (0.4, 0.2, 0.4, 0)$ is stationary for $\mathbf{H}$: a page nobody links to scores 0.

Communicating Classes and Irreducibility

Definition

State i **leads to** j , written $i \rightarrow j$, if $p^{(n)}(i, j) > 0$ for some $n \geq 0$. States i, j **communicate**, $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$. This is an equivalence relation on S ; its equivalence classes are the **communicating classes**. A class C is **closed** if $i \in C$ and $i \rightarrow j$ imply $j \in C$. The chain is **irreducible** if S is a single class.

Two states are in the same class if the chain can get from each to the other; a closed class is one the chain cannot leave; irreducible means every state is reachable from every other.

- $i \rightarrow j$ iff there is a path $i = i_0, i_1, \dots, i_n = j$ with every $p(i_{k-1}, i_k) > 0$: a directed path in the graph whose edges are the positive entries of $\mathbf{P}$. The classes are the strongly connected components of that graph.
- Weather, vowel/consonant, Ehrenfest, cycle, cube, PageRank with $d < 1$: irreducible.
- Gambler's ruin: three classes, $\{0\}$ and $\{N\}$ closed (absorbing states), $\{1, \dots, N-1\}$ not closed. Four-page web without damping: $\{A, B, C\}$ closed, $\{D\}$ not closed.
- A state in a class that is not closed is left for good with positive probability at each visit, so it is visited only finitely often (next slides).

Periodicity

Definition

The **period** of state i is $d(i) = \gcd\{n \geq 1 : p^{(n)}(i, i) > 0\}$ (with $d(i) = \infty$ if the set is empty). State i is **aperiodic** if $d(i) = 1$. A chain is aperiodic if every state is.

The period is the common divisor of all the lengths of round trips from i ; period 2 means every return takes an even number of steps.

Theorem

All states in a communicating class have the same period. If $p(i, i) > 0$ for some i in a class, the class is aperiodic.

Proof

Let $i \leftrightarrow j$ with $p^{(a)}(i, j) > 0$, $p^{(b)}(j, i) > 0$. Every round trip from j of length n gives round trips from i of lengths $a + b$ and $a + n + b$, so $d(i)$ divides both, hence divides n ; thus $d(i) \leq d(j)$, and symmetrically. If $p(i, i) > 0$ then 1 is a round-trip length. □

Periodicity: The Cube Never Forgets Its Parity

Random walk on the cube

Each step flips one coordinate, so the parity of the number of 1s alternates: even at even times, odd at odd times. Hence $p^{(n)}(000, 000) = 0$ for odd n , $d = 2$, and

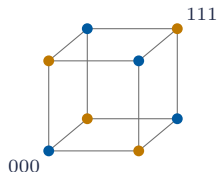
$$(\mathbf{P}^{10})_{000, \cdot} \approx \frac{1}{4}(1, 0, 0, 1, 0, 1, 1, 0),$$

$$(\mathbf{P}^{11})_{000, \cdot} \approx \frac{1}{4}(0, 1, 1, 0, 1, 0, 0, 1)$$

(order 000, 100, 010, 110, 001, 101, 011, 111; zeros exact; $p^{(10)}(000, 000) = \frac{1}{4} + \frac{3}{4}3^{-10}$). The rows of $\mathbf{P}^n$ do not converge: they alternate between two limits, while $\pi = (\frac{1}{8}, \dots, \frac{1}{8})$ is stationary all along.

A memory of the starting time that never fades: π exists, but the chain never settles into it.

- **Cycle** of length L : $d = 2$ if L is even, $d = 1$ if L is odd (round trips of lengths 2 and L).
Ehrenfest: $d = 2$. **Weather, vowel/consonant**: $p(i, i) > 0$, so $d = 1$.
- Cure for periodicity: a lazy version, $\mathbf{P}' = \frac{1}{2}(\mathbf{I} + \mathbf{P})$, stays put with probability $\frac{1}{2}$, has the same π , and $d = 1$.



Blue: even parity; amber: odd. Every edge joins the two colours, so the walk alternates colours forever.

Recurrence and Transience

Definition

The **return time** to i is $T_i: \Omega \rightarrow \{1, 2, \dots\} \cup \{\infty\}$, $T_i = \min\{n \geq 1 : X_n = i\}$ (∞ if there is no such n). State i is **recurrent** if $\mathbb{P}_i(T_i < \infty) = 1$ and **transient** if $\mathbb{P}_i(T_i < \infty) < 1$. A recurrent state is **positive recurrent** if $\mathbb{E}_i[T_i] < \infty$ and **null recurrent** otherwise.

Recurrent: started at i , the chain is certain to come back; transient: with positive probability it never does.

- T_i is a row quantity: the first column after 0 in which the path shows i . It is the waiting time of the Conditional Expectation lecture, and for the gambler it was the duration of the game.
- Gambler's ruin: 0 and N are recurrent ($T_i = 1$ surely); $1, \dots, N - 1$ are transient, since the game ends with probability one and then never returns.
- Random walk on $\mathbb{Z}$: state 0 is recurrent iff $p = \frac{1}{2}$ (slide 60). Recurrent and yet $\mathbb{E}_0[T_0] = \infty$: the fair walk is null recurrent, the sharp form of "ruin certain but $\mathbb{E}[T] = \infty$ " on slide 33.
- Positive recurrence is what makes long-run frequencies meaningful: returns every $\mathbb{E}_i[T_i]$ steps on average, so a fraction $1/\mathbb{E}_i[T_i]$ of time is spent at i (slide 72).

Recurrence: The Visits Criterion

Theorem

Let $V_i: \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$, $V_i = \sum_{n \geq 1} \mathbf{1}\{X_n = i\}$, be the number of returns to i , and $f = \mathbb{P}_i(T_i < \infty)$. Then

$$\mathbb{P}_i(V_i \geq k) = f^k, \quad \mathbb{E}_i[V_i] = \sum_{n \geq 1} p^{(n)}(i, i) = \frac{f}{1-f} \quad (= \infty \text{ if } f = 1).$$

Hence i is recurrent iff $\sum_n p^{(n)}(i, i) = \infty$ iff $\mathbb{P}_i(V_i = \infty) = 1$; transient iff $\sum_n p^{(n)}(i, i) < \infty$ iff $V_i < \infty$ with probability one.

Proof

Each time the chain is at i it restarts (Markov property), and returns once more with probability f , independently of the past; so $V_i \geq k$ requires k successive returns: f^k . Then

$\mathbb{E}_i[V_i] = \sum_{k \geq 1} \mathbb{P}_i(V_i \geq k) = \sum_k f^k$, and also $\mathbb{E}_i[V_i] = \sum_n \mathbb{E}_i[\mathbf{1}\{X_n = i\}] = \sum_n p^{(n)}(i, i)$ by linearity, column by column. □

Recurrence: Class Property, Finite Chains, and the Walk on $\mathbb{Z}$

Theorem

- (a) Recurrence and transience are class properties: all states of a class are recurrent, or all are transient.
- (b) Every finite closed class is recurrent; every class that is not closed is transient. In particular a finite irreducible chain is recurrent, indeed positive recurrent, and its stationary distribution is unique with $\pi_i = 1/\mathbb{E}_i[T_i]$.

In a finite chain the only way to avoid returning is to leave the class for good; inside a closed class the chain has nowhere else to go, and it returns on average every $1/\pi_i$ steps.

Proof sketch

(a) If $i \leftrightarrow j$ with $p^{(a)}(i, j)p^{(b)}(j, i) > 0$, then $p^{(a+n+b)}(i, i) \geq p^{(a)}(i, j)p^{(n)}(j, j)p^{(b)}(j, i)$, so the two series $\sum_n p^{(n)}(i, i)$, $\sum_n p^{(n)}(j, j)$ diverge together. (b) Started in a finite closed class, some state is visited infinitely often, so that state is recurrent, so all are by (a). Not closed: leaving is possible and permanent. The formula $\pi_i = 1/\mathbb{E}_i[T_i]$ is the ergodic theorem, slide 72. $\square$

- Weather: mean return time to R is $1/\pi_R = \frac{13}{3}$ days. Vowel/consonant: mean return to V is $1/\pi_V = \frac{13}{5}$, which first-step analysis confirms: $1 + 0.8 \cdot 2$, since from C the wait for V is geometric with mean 2.

The Random Walk on $\mathbb{Z}$ Is Recurrent If and Only If It Is Fair

Theorem

The simple random walk on $\mathbb{Z}$ is recurrent if $p = \frac{1}{2}$ (null recurrent) and transient if $p \neq \frac{1}{2}$.

Without drift the walk keeps coming home, at ever longer intervals; with any drift it eventually leaves for good.

Proof

Returns to 0 take an even number of steps, n up and n down: $p^{(2n)}(0,0) = \binom{2n}{n} p^n q^n$ (slide 7). By Stirling's formula (not proved here) $\binom{2n}{n} \approx 4^n / \sqrt{\pi n}$, the ratio tending to 1, so

$$p^{(2n)}(0,0) \approx \frac{(4pq)^n}{\sqrt{\pi n}}.$$

Fair: $4pq = 1$ and $\sum_n (\pi n)^{-1/2} = \infty$: 0 is recurrent by the visits criterion. Unfair: $4pq = 1 - (p - q)^2 < 1$, the series converges geometrically: transient. Null recurrence: $\mathbb{E}_0[T_0] = \infty$ by slide 33. $\square$

- $p = 0.6$: $4pq = 0.96$ and $\mathbb{E}_0[V_0] = \sum_{n \geq 1} \binom{2n}{n} (pq)^n = (1 - 4pq)^{-1/2} - 1 = 4$, so $f = \frac{4}{5} = 1 - |p - q|$ by the visits criterion: four returns on average, then gone for good.

Hitting Probabilities: The Minimal Solution of a First-Step System

Definition

For $A \subseteq S$, the **hitting time** of A is $T_A: \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$, $T_A = \min\{n \geq 0 : X_n \in A\}$, and the **hitting probability** is the function

$$h^A: S \rightarrow [0, 1], \quad h_i^A = \mathbb{P}_i(T_A < \infty).$$

Theorem

h^A is the **minimal non-negative solution** of the system

$$h_i = 1 \quad (i \in A), \quad h_i = \sum_{j \in S} p(i, j) h_j \quad (i \notin A):$$

it solves it, and every solution x with all $x_i \geq 0$ has $x_i \geq h_i^A$ for all i .

Off the target the hitting probability is the row average of itself, one step ahead; if that system has several non-negative solutions, the smallest is the right one.

- A is the fixed target set; i is the starting state, the variable of h^A . Gambler's ruin:
 $S = \{0, \dots, N\}$, $A = \{N\}$ (next slide); the exit problem of slide 11: $A = \{-a, a\}$.

Hitting Probabilities: Proof

Proof

Solves it: $h_i^A = 1$ on A ; off A , condition on the first step (total probability over the rows $\{X_1 = j\}$) and use the Markov property: $\mathbb{P}_i(T_A < \infty) = \sum_j p(i, j) \mathbb{P}_j(T_A < \infty)$. Minimal: let $x \geq 0$ solve the system. Then $x_i \geq \mathbb{P}_i(T_A \leq n)$ for all n , by induction: true for $n = 0$ (on A , $x_i = 1$; off A , $x_i \geq 0 = \mathbb{P}_i(T_A \leq 0)$); and for $i \notin A$,

$$x_i = \sum_j p(i, j) x_j \geq \sum_j p(i, j) \mathbb{P}_j(T_A \leq n) = \mathbb{P}_i(T_A \leq n + 1)$$

by the induction hypothesis and first-step analysis. Let $n \rightarrow \infty$: $x_i \geq \mathbb{P}_i(T_A < \infty) = h_i^A$. □

Any non-negative solution dominates the probability of hitting within n steps for every n , hence dominates the probability of ever hitting.

- On a finite state space with A reachable from everywhere, the system has a unique solution and minimality is automatic; at absorbing states outside A , and on infinite state spaces, it is what selects the answer.

Hitting Probabilities: The Gambler's Ruin Is the Case $A = \{N\}$

Finite ruin chain, $A = \{N\}$

The system reads $h_N = 1$; $h_i = p h_{i+1} + q h_{i-1}$ for $0 < i < N$; and at the absorbing state 0: $h_0 = \sum_j p(0, j) h_j = h_0$, which says nothing. The general solution is $h_i = a + b r^i$ (or $a + bi$ if $p = q$) for $i \geq 1$ with h_0 free; non-negative solutions exist with any $h_0 \in [0, 1]$, and the *minimal* one has $h_0 = 0$, giving slides 28–29.

The boundary condition $h_0 = 0$ that we imposed by hand on slide 27 is what minimality supplies automatically: an absorbing state you cannot leave has hitting probability 0 for any target that is not itself.

- Every hitting probability of this lecture is one system: $A = \{N\}$ gives the chance of reaching the target, $A = \{0\}$ the chance of ruin, and $A = \{0, N\}$ gives $h \equiv 1$ (the game ends).
- Conversely, the ruin argument (condition on the first round, restart) is the proof that h^A solves the system; nothing new was needed for general chains except minimality.

Why “Minimal”: The Infinite Ruin Chain

Infinite ruin chain, $S = \{0, 1, 2, \dots\}$, $A = \{0\}$ (slide 32 again)

$h_0 = 1$ and $h_i = p h_{i+1} + q h_{i-1}$ for $i \geq 1$; general solution $h_i = 1 + b(r^i - 1)$ with $r = q/p$.

- $p > q$ ($r < 1$): $h_i \geq 0$ for all i requires $b \leq 1$, and h_i decreases as b increases, so the minimal solution is $b = 1$: $h_i = r^i = (q/p)^i$.
- $p < q$ ($r > 1$): $h_i \geq 0$ for all i requires $b \geq 0$ (otherwise $h_i \rightarrow -\infty$), minimal at $b = 0$: $h_i = 1$.
- $p = q$: $h_i = 1 + bi$, $b \geq 0$, minimal $h_i = 1$.

$h \equiv 1$ solves the system in every case; it is the answer only when it is the smallest non-negative solution.

The system alone cannot tell “ruin certain” from “ruin probability $(q/p)^i$ ”; minimality does, and it agrees with the limit $N \rightarrow \infty$ of slide 32.

- The same happens for transient chains generally: on an infinite state space, a non-negative solution can be inflated by adding a positive solution of the homogeneous equation that vanishes on A ; only the smallest is a probability.

Expected Hitting Times

Theorem

Let $k^A: S \rightarrow [0, \infty]$, $k_i^A = \mathbb{E}_i[T_A]$. Then k^A is the minimal non-negative solution of

$$k_i = 0 \quad (i \in A), \quad k_i = 1 + \sum_{j \in S} p(i, j) k_j \quad (i \notin A).$$

Off the target, pay one step and add the row average of the remaining expected times; the exit-time and duration equations of slides 11 and 30 are this system for the walk.

Proof

Solves it: off A , $T_A = 1 + T'_A$ where T'_A is the hitting time of the chain restarted at X_1 ; take $\mathbb{E}_i$ and condition on X_1 . Minimal: as before, any non-negative solution x satisfies $x_i \geq \mathbb{E}_i[\min(T_A, n)]$ for all n , by induction. $\square$

- The system may have no finite non-negative solution at all; then $k_i^A = \infty$ for some i , which is what the fair walk on $\mathbb{Z}$ does. A finite solution, when the target is reachable from everywhere in a finite chain, is unique.
- In matrix form on a finite chain: $(\mathbf{I} - \mathbf{Q})k = \mathbf{1}$ with $\mathbf{Q}$ the transitions among the states off A ; slide 75 inverts it.

Expected Hitting Times: Corner to Opposite Corner on the Cube

How many steps does the random walk on the cube take, on average, to get from 000 to 111?

Head-on

$\mathbb{E}_{000}[T_{111}] = \sum_n n \mathbb{P}_{000}(T_{111} = n)$, a sum over paths on 8 states of every length that avoid 111 until the end. No column of the table answers it.

Random walk on the cube: from 000 to the opposite corner 111

Group states by distance $d \in \{0, 1, 2, 3\}$ from 111; by symmetry k depends only on d . From distance d a step goes to distance $d - 1$ with probability $d/3$ and to $d + 1$ with probability $(3 - d)/3$:

$$k_0 = 0, \quad k_1 = 1 + \frac{1}{3}k_0 + \frac{2}{3}k_2, \quad k_2 = 1 + \frac{2}{3}k_1 + \frac{1}{3}k_3, \quad k_3 = 1 + k_2,$$

giving $k_1 = 7$, $k_2 = 9$, $k_3 = 10$: ten steps on average from a corner to the opposite corner.

Symmetry collapsed eight unknowns to three, and first-step analysis turned an infinite sum over paths into three linear equations.

- Check: from a neighbour of 000 the expected time back to 000 is 7, so the mean return time to 000 is $1 + 7 = 8 = 1/\pi_{000}$, as the ergodic theorem predicts (slide 72).

Return to Question 3: Days Until Rain

$A = \{R\}$, $k_R = 0$, and for the other two states

$$k_S = 1 + 0.7 k_S + 0.2 k_C, \quad k_C = 1 + 0.3 k_S + 0.4 k_C.$$

Answer

From the second, $0.6 k_C = 1 + 0.3 k_S$; substituting into the first, $0.3 k_S = 1 + 0.2 k_C$, so $k_S = \frac{20}{3} \approx 6.7$ days and $k_C = 5$ days.

Consistency check with the stationary distribution

The mean return time to R is one step plus the expected hitting time from wherever that step lands:

$$\mathbb{E}_R[T_R] = 1 + 0.2 k_S + 0.4 k_C + 0.4 \cdot 0 = 1 + \frac{4}{3} + 2 = \frac{13}{3} = \frac{1}{\pi_R}.$$

The three opening questions have each been answered by one operation on the transition table: a power, a left eigenvector, and a first-step linear system.

- Vowel/consonant, the same tool: expected letters until the first vowel is 2 from C (geometric) and $1 + 0.8 \cdot 2 = 2.6$ from V. Until the pattern VV: $k_C = 15$, $k_V = 13$.

The Convergence Theorem

Theorem

Let $\mathbf{P}$ be irreducible and aperiodic on a finite state space S . Then $\mathbf{P}$ has a unique stationary distribution π , and for all $i, j \in S$,

$$p^{(n)}(i, j) \xrightarrow{n \rightarrow \infty} \pi_j.$$

Equivalently, for every initial distribution λ , $\lambda \mathbf{P}^n \rightarrow \pi$: every row of $\mathbf{P}^n$ converges to π .

Run the chain long enough and where it is has almost the distribution π , whatever the starting point: the chain forgets where it began.

- Slide 46: rows of the vowel/consonant $\mathbf{P}^n$ approaching $(\frac{5}{13}, \frac{8}{13})$. Slide 45: rows of the weather $\mathbf{P}^7$ agreeing to two places. PageRank: $\mathbf{P}^{30}$ has all rows equal to π to four places, which is how π is computed for 10^{10} pages.
- The theorem is about *columns* of the table of paths: the distribution at a late time. It says nothing about the path itself, which keeps moving; the ergodic theorem (slide 72) is the row statement.
- Two hypotheses, both necessary: irreducibility for uniqueness of π and independence of the start; aperiodicity for the convergence itself. Next slide.

Why Each Hypothesis Is Needed

Reducible: the limit depends on the start

Two closed classes, $\mathbf{P} = \begin{pmatrix} \mathbf{P}_1 & 0 \\ 0 & \mathbf{P}_2 \end{pmatrix}$. Rows of $\mathbf{P}^n$ starting in class 1 converge to $(\pi_1, 0)$, in class 2 to $(0, \pi_2)$; every mixture $(\alpha\pi_1, (1-\alpha)\pi_2)$ is stationary. Gambler's ruin: rows converge to $(1 - h_i, 0, \dots, 0, h_i)$, different for each i ; for $N = 4$, $p = \frac{1}{2}$, row i tends to $(1 - \frac{i}{4}, 0, 0, 0, \frac{i}{4})$.

Periodic: the rows do not converge

The cube (slide 56): rows of $\mathbf{P}^n$ alternate between two vectors, though $\pi = (\frac{1}{8}, \dots, \frac{1}{8})$ is unique. Likewise the Ehrenfest urn and the cycle of even length.

Infinite $\mathcal{S}$: there may be nothing to converge to

The fair walk on $\mathbb{Z}$ is irreducible, but has no stationary distribution (slide 48) and $p^{(2n)}(0, 0) \approx 1/\sqrt{\pi n} \rightarrow 0$ (slide 60): every entry of every row of $\mathbf{P}^n$ tends to 0, and the mass spreads out over ever more states.

Reducible: which limit depends on where you start; periodic: no limit at all; infinite: the limit may be zero everywhere.

How Fast: The Second Eigenvalue

Theorem (rate of convergence, diagonalisable case)

Suppose $\mathbf{P}$ is diagonalisable, with eigenvalues $1 = \lambda_1, \lambda_2, \dots, \lambda_{|S|}$, right eigenvectors $\mathbf{v}_k$ (columns, $\mathbf{P}\mathbf{v}_k = \lambda_k \mathbf{v}_k$) and left eigenvectors $\mathbf{u}_k$ (rows, $\mathbf{u}_k \mathbf{P} = \lambda_k \mathbf{u}_k$), chosen so that $\mathbf{u}_k \mathbf{v}_k = 1$ and $\mathbf{u}_k \mathbf{v}_l = 0$ for $k \neq l$. Then $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{u}_1 = \pi$, $\mathbf{P} = \sum_k \lambda_k \mathbf{v}_k \mathbf{u}_k$, and

$$\mathbf{P}^n = \mathbf{1}\pi + \sum_{k \geq 2} \lambda_k^n \mathbf{v}_k \mathbf{u}_k, \quad |p^{(n)}(i, j) - \pi_j| \leq C |\lambda_2|^n, \quad |\lambda_2| = \max_{k \geq 2} |\lambda_k| < 1.$$

Every term of $\mathbf{P}^n$ after the limit $\mathbf{1}\pi$ carries a factor of modulus below one, so the error shrinks geometrically at the rate of the second-largest eigenvalue.

Proof

$\mathbf{P}\mathbf{1} = \mathbf{1}$ (row sums) and $\pi\mathbf{P} = \pi$, $\pi\mathbf{1} = 1$ give $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{u}_1 = \pi$. As $(\mathbf{v}_k \mathbf{u}_k)(\mathbf{v}_l \mathbf{u}_l) = (\mathbf{u}_k \mathbf{v}_l) \mathbf{v}_k \mathbf{u}_l$ vanishes for $k \neq l$, powers act termwise: $(\sum_k \lambda_k \mathbf{v}_k \mathbf{u}_k)^n = \sum_k \lambda_k^n \mathbf{v}_k \mathbf{u}_k$. Irreducibility rules out a second eigenvalue 1 (a second stationary distribution), aperiodicity the other eigenvalues of modulus 1 (the cube has -1); these two facts are not proved here. $\square$

- Vowel/consonant: $\lambda_2 = -0.3$, error $\frac{8}{13}(0.3)^n$ (slide 46). Weather: eigenvalues 1, 0.456, 0.044; $\max_{i,j} |p^{(n)}(i, j) - \pi_j| = 0.26, 0.056, 0.0053, 0.0024$ at $n = 1, 3, 6, 7$.

Proof Sketch by Coupling

Idea

Run two independent copies of the chain: X started at i , and Y started from π , so that $Y_n \sim \pi$ for every n (slide 47). Let $\tau = \min\{n : X_n = Y_n\}$ be the first time they meet. Define $Z_n = X_n$ for $n < \tau$ and $Z_n = Y_n$ for $n \geq \tau$: switch to following Y once they coincide.

Two facts

1. Z is again a Markov chain with matrix $\mathbf{P}$ started at i (at the switch, both copies are at the same state and both continue by the same rule), so $\mathbb{P}(Z_n = j) = p^{(n)}(i, j)$.
2. $Z_n = Y_n$ whenever $\tau \leq n$, so $|p^{(n)}(i, j) - \pi_j| = |\mathbb{P}(Z_n = j) - \mathbb{P}(Y_n = j)| \leq \mathbb{P}(\tau > n)$.

Where the hypotheses enter

The pair (X_n, Y_n) is a Markov chain on $S \times S$ with transitions $p(i, i')p(j, j')$. Irreducibility and aperiodicity of $\mathbf{P}$ make this pair chain irreducible (aperiodicity is needed: for the cube, a pair at different parities can never meet), hence recurrent on the finite $S \times S$, so it reaches the diagonal: $\mathbb{P}(\tau < \infty) = 1$, i.e. $\mathbb{P}(\tau > n) \rightarrow 0$. □

The chain from i and the chain from π are indistinguishable after their first meeting, and they are certain to meet.

The Ergodic Theorem: Long-Run Fractions of Time

Theorem

Let $\mathbf{P}$ be irreducible on a finite S with stationary distribution π (no aperiodicity required). Let $V_i(n): \Omega \rightarrow \{0, \dots, n\}$, $V_i(n) = \sum_{k=0}^{n-1} \mathbf{1}\{X_k = i\}$, be the number of visits to i before time n (time 0 included, unlike V_i of slide 58: $V_i(n) \rightarrow V_i + \mathbf{1}\{X_0 = i\}$). Then, whatever the initial distribution, with probability one

$$\frac{V_i(n)}{n} \xrightarrow{n \rightarrow \infty} \pi_i \quad \text{for every } i, \quad \text{and} \quad \frac{1}{n} \sum_{k=0}^{n-1} f(X_k) \rightarrow \sum_{i \in S} \pi_i f(i) \quad \text{for every } f: S \rightarrow \mathbb{R}.$$

Moreover $\pi_i = 1/\mathbb{E}_i[T_i]$.

Along a single path, the fraction of time spent in i settles to π_i , and time averages of any function equal its π -average.

- $V_i(n)/n$ is a *row* quantity: read one path of the table and count. The theorem says almost every row has the same limiting frequencies, and they are π .
- f is a reward per state: long-run average reward equals the π -weighted reward. With $f = \mathbf{1}_{\{i\}}$ this is the first statement.

The Ergodic Theorem: Proof Idea and a Time Average

Proof idea

The visits to i split the path into independent, identically distributed excursions of mean length $\mathbb{E}_i[T_i]$ (Markov property at each return). By the law of large numbers (LLN: averages of i.i.d. variables with finite mean converge to that mean with probability one), m excursions take about $m \mathbb{E}_i[T_i]$ steps, so in n steps there are about $n/\mathbb{E}_i[T_i]$ visits: $V_i(n)/n \rightarrow \mu_i := 1/\mathbb{E}_i[T_i]$.

Why the limit is π

Among the visits to i , a fraction $p(i, j)$ is followed by a step to j (Markov property, LLN), so the fraction of times $k < n$ with $X_k = i$, $X_{k+1} = j$ tends to $\mu_i p(i, j)$. Summing over i counts the visits to j at times $1, \dots, n$, whose fraction tends to μ_j : $\mu_j = \sum_i \mu_i p(i, j)$, i.e. $\mu \mathbf{P} = \mu$, so $\mu = \pi$ by uniqueness (slide 59). □

Long-run frequencies balance because every arrival at j is a departure from some i .

The cube: a time average although the rows of $\mathbf{P}^n$ never converge

$f(x)$ = number of 1s in x . Along any path, with probability one,
 $\frac{1}{n} \sum_{k < n} f(X_k) \rightarrow \sum_i \pi_i f(i) = \frac{1}{8}(0 + 3 \cdot 1 + 3 \cdot 2 + 3) = \frac{3}{2}$, and the fraction of time at even-parity corners tends to $\frac{1}{2}$, although at each fixed time n that probability is 1 or 0 (slide 56).

Return to Question 2, and the Two Theorems Side by Side

Weather, question 2

In the long run a fraction $\pi_S = \frac{6}{13} \approx 0.46$ of days are sunny, $\frac{4}{13}$ cloudy, $\frac{3}{13}$ rainy: with probability one, along the actual sequence of days. And the fraction of days that are rainy *and* followed by a sunny day is $\pi_R p(R, S) = \frac{3}{13} \cdot 0.2 = \frac{3}{65} \approx 0.046$ (apply the theorem to the pair chain (X_n, X_{n+1})).

	convergence theorem	ergodic theorem
statement	$\mathbb{P}(X_n = i) \rightarrow \pi_i$	$V_i(n)/n \rightarrow \pi_i$ with probability one
reads the table	down a column (late time)	along a row (one path)
needs	irreducible, aperiodic, finite	irreducible, finite
fails for the cube?	yes: rows of $\mathbf{P}^n$ alternate	no: $\frac{1}{8}$ of the time at each corner
what π_i means	where the chain is, eventually	how often the chain is there, on average
computes	$\mathbf{P}^n$ for large n	π from $\pi \mathbf{P} = \pi$, or $1/\mathbb{E}_i[T_i]$

Both theorems say π is the long-run behaviour of the chain; the first in distribution at a fixed late time, the second in frequency along a single path.

Absorbing Chains and the Fundamental Matrix

Definition

A chain is **absorbing** if every state leads to some absorbing state. Order the states transient first, absorbing last: with S_T the transient and S_A the absorbing states,

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{R} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}, \quad \mathbf{Q}: S_T \times S_T \rightarrow [0, 1], \quad \mathbf{R}: S_T \times S_A \rightarrow [0, 1].$$

The **fundamental matrix** is $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}: S_T \times S_T \rightarrow [0, \infty)$.

Theorem

$\mathbf{I} - \mathbf{Q}$ is invertible and $\mathbf{N} = \sum_{n \geq 0} \mathbf{Q}^n$. For transient i, j and absorbing a :

$$\mathbf{N}_{ij} = \mathbb{E}_i[\#\{n \geq 0 : X_n = j\}], \quad (\mathbf{N}\mathbf{1})_i = \mathbb{E}_i[T_{S_A}], \quad (\mathbf{N}\mathbf{R})_{ia} = \mathbb{P}_i(\text{absorbed at } a).$$

$\mathbf{Q}^n$ is the table of n -step transitions that avoid absorption, so summing over n counts expected visits; row sums are expected times to absorption, and one more step into $\mathbf{R}$ gives where.

The Fundamental Matrix: Proof Sketch

Proof sketch

$(Q^n)_{ij} = \mathbb{P}_i(X_n = j)$ for transient j (paths through transient states only), so $\sum_n (Q^n)_{ij} = \mathbb{E}_i[\text{visits to } j]$ by linearity, column by column. Absorption is certain and from a finite set of transient states happens geometrically fast, so $Q^n \rightarrow 0$ and the series converges, to $(I - Q)^{-1}$ since $(I - Q) \sum_{n < m} Q^n = I - Q^m$. Time to absorption is the total number of transient visits: $\sum_j N_{ij}$. Absorption at a : visit some j then step to a : $\sum_j N_{ij} R_{ja}$. $\square$

Three quantities, one inverse: expected visits, expected time, and where the chain ends up all come from $(I - Q)^{-1}$.

- The hitting systems of slides 61 and 65 are $(I - Q)k = \mathbf{1}$ and $(I - Q)h = R_{\cdot, a}$; the fundamental matrix is their solution operator.
- Any chain becomes absorbing for the purpose of a hitting question: make the target states absorbing (replace their rows by point masses), which does not change the path up to the hitting time.

The Fundamental Matrix of the Ruin Chain, $N = 4$

Transient states 1, 2, 3; absorbing 0, 4. Fair case:

$$\mathbf{Q} = \begin{pmatrix} 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} = \begin{pmatrix} \frac{3}{2} & 1 & \frac{1}{2} \\ 1 & 2 & 1 \\ \frac{1}{2} & 1 & \frac{3}{2} \end{pmatrix},$$

$$\mathbf{N}\mathbf{1} = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}, \quad \mathbf{N}\mathbf{R} = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

Reading it

$\mathbf{N}\mathbf{1} = (3, 4, 3) = i(4 - i)$: the durations of slide 30. Column 4 of $\mathbf{N}\mathbf{R}$ is $(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}) = i/4$: slide 28.

New information: $\mathbf{N}_{2,2} = 2$ visits to fortune 2 (counting the start) before the game ends when starting there, and $\mathbf{N}_{1,3} = \frac{1}{2}$ visits to 3 when starting from 1.

- $p = \frac{2}{5}$: $\mathbf{N}\mathbf{1} = (\frac{33}{13}, \frac{50}{13}, \frac{43}{13})$ and the target column of $\mathbf{N}\mathbf{R}$ is $(\frac{8}{65}, \frac{4}{13}, \frac{38}{65})$, agreeing with the closed forms of slides 29–30. One matrix inverse replaces both derivations and any boundary-condition bookkeeping.

Summary

object	type	one-glance meaning in the table
process $(X_t)_{t \in T}$	$X_t: \Omega \rightarrow \mathbb{R}$ for each t	columns are random variables, rows are paths
random walk S_n	$\Omega \rightarrow \mathbb{Z}$	sum of ± 1 steps; $\text{Var}(S_n) = 4npq$ ($= n$ if fair); restarts after each step
Brownian motion B_t	$[0, \infty) \times \Omega \rightarrow \mathbb{R}$	the walk at $\sqrt{n}$ scale; independent $N(0, t - s)$ increments
ruin probability $1 - h_i$	$\{0, \dots, N\} \rightarrow [0, 1]$	$1 - i/N$, or $\frac{r^N - r^i}{r^N - 1}$; from $h_i = ph_{i+1} + qh_{i-1}$
transition matrix $\mathbf{P}$	$S \times S \rightarrow [0, 1]$, rows sum to 1	row i is the conditional pmf of X_{n+1} given $X_n = i$
n -step $\mathbf{P}^n$, law $\lambda \mathbf{P}^n$	matrix, row vector	Chapman–Kolmogorov: condition on the middle state
stationary π	$S \rightarrow [0, 1]$, $\pi \mathbf{P} = \pi$	inflow equals outflow at every state; detailed balance if reversible
classes, period, recurrence	structure of $\mathbf{P}$'s graph	who reaches whom, in steps of what length, and returns surely or not
hitting h^A, k^A	$S \rightarrow [0, 1]$, $S \rightarrow [0, \infty]$	minimal solutions of first-step systems; NR , N1
convergence, ergodic	$\mathbf{P}^n \rightarrow \mathbf{1}\pi$; $V_i(n)/n \rightarrow \pi_i$	the column limit and the row limit are both π

One tool did most of the work: condition on the first step, and a question about paths of unbounded length

becomes a finite linear system.